

Anàlisi de dades òmiques (M0-157). PAC1.

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1. Introducció

Aquest arxiu conté les sol·lucions analítiques als problemes plantejats en la PAC1 *Anàlisi de dades òmiques (M0-157)*. El projecte total del treball es pot consultar al: [meu repositori github](#), cliqueu **aquí**

2. Carrega de dades

Abans que res, carreguem les dades al nostre entorn de treball. Treballarem amb un dels exemples de mostra, veure [aquí](#). Aquest exemple porta dos arxius, DataInfo_S013 i DataValues_S013. Carreguem els dos fitxers:

```
DataInfo_S013 <- read.csv("DataInfo_S013.csv", sep = ",")
DataValues_S013 <- read.csv("DataValues_S013.csv", sep = ",")
```

Com és habitual amb dades òmiques, tenim moltes varibales i un nombre reduït de casos d'estudi. Tot i així, abans que res mirem l'estructura dels dos.

3. Exploració previa de les dades descarregades

```
dim(DataInfo_S013)
```

```
## [1] 695 4
```

```
dim(DataValues_S013)
```

```
## [1] 39 696
```

Aquí ja veiem que en l'arxiu Info hi ha 695 files que han de correspondre a les variables que hi ha a Values. En aquest darrer tenim 696 columnes i 39 files. Les files corresponen a cada cas d'estudi més les 695 columnes són les variables que es tenen de cada un d'ells.

Per veure què són les variables, mirem les 20 primeres files d'Info i podem fer-nos una idea.

```
head(DataInfo_S013,20)
```

##	X	VarName	varTpe	Description
## 1	SUBJECTS	SUBJECTS	integer	dataDesc
## 2	SURGERY	SURGERY	character	dataDesc
## 3	AGE	AGE	integer	dataDesc
## 4	GENDER	GENDER	character	dataDesc
## 5	Group	Group	integer	dataDesc
## 6	MEDDM_TO	MEDDM_TO	integer	dataDesc
## 7	MEDCOL_TO	MEDCOL_TO	integer	dataDesc
## 8	MEDINF_TO	MEDINF_TO	integer	dataDesc
## 9	MEDHTA_TO	MEDHTA_TO	integer	dataDesc

```
## 10      GLU_TO      GLU_TO      integer      dataDesc
## 11      INS_TO      INS_TO      numeric      dataDesc
## 12      HOMA_TO      HOMA_TO      numeric      dataDesc
## 13      HBA1C_TO      HBA1C_TO      numeric      dataDesc
## 14 HBA1C.mmol.mol_TO HBA1C.mmol.mol_TO numeric      dataDesc
## 15      PESO_TO      PESO_TO      integer      dataDesc
## 16      bmi_TO      bmi_TO      numeric      dataDesc
## 17      CC_TO      CC_TO      numeric      dataDesc
## 18      CINT_TO      CINT_TO      integer      dataDesc
## 19      CAD_TO      CAD_TO      integer      dataDesc
## 20      TAD_TO      TAD_TO      integer      dataDesc
```

Aquí ja veiem que hi ha 9 primers camps que semblen ser els que tenen la informació sobre el pacient de cada mostra. Mirem que contenen aquests camps a values:

```
# En mirem 10 en realitat, per comprovar que la primera columna és un index
head(DataValues_S013[,1:10])
```

```
##      X.1 SUBJECTS SURGERY AGE GENDER Group MEDDM_TO MEDCOL_TO MEDINF_TO MEDHTA_TO
## 1      1          1 by pass 27      F      1          0          0          0          1
## 2      2          2 by pass 19      F      2          0          0          0          0
## 3      3          3 by pass 42      F      1          0          0          0          0
## 4      4          4 by pass 37      F      2          0          0          0          0
## 5      5          5 tubular 42      F      1          0          0          0          0
## 6      6          6 by pass 24      F      2          0          0          0          0
```

Com suposavem, aquestes columnes corresponen a: 1. Un codi de pacient (que està repetit a les dues primeres columnes, d'aquí que hi hagi una fila més a Info que columnes a Values) 2. La cirurgia a la que va ser sotmès el pacient 3. L'edat i el gènere 4. Un grup de tractament 5. 4 tractaments que venen amb 0 i 1

Per veure que la resta són variables metabòliques, mirem 10 columnes més

```
head(DataValues_S013[,11:20],10)
```

```
##      GLU_TO INS_TO HOMA_TO HBA1C_TO HBA1C.mmol.mol_TO PESO_TO bmi_TO CC_TO
## 1      85  11.40   2.40      NA          NA      151  62.9   0.7
## 2      78  12.10   2.32      NA          NA      139  47.0   NA
## 3      75   8.41   1.56    5.4      35.51    84  29.8   0.7
## 4      71  12.80   2.25    5.1      32.23   136  53.1   1.0
## 5      82   6.01   1.22    5.6      37.69   121  46.6   0.9
## 6      71   9.88   1.73    5.1      32.23   148  48.8   0.7
## 7      80   9.20   1.82    5.6      37.69   109  43.7   0.9
## 8      90   3.40   0.76    5.5      36.60   109  41.8   0.9
## 9      92   5.43   1.23    5.7      38.78   114  44.0   0.9
## 10     84   6.98   1.45    5.5      36.60   120  40.6   NA
##      CINT_TO CAD_TO
## 1      116    167
## 2      NA     NA
## 3      90    126
## 4     157    162
## 5     123    132
## 6     110    148
## 7     122    141
## 8     124    136
## 9     136    148
## 10     NA     NA
```

Obviament ja ho veiem i també podem veure que hi ha més d'un valor que no es té per a alguna de les

variables (NA). Anem a veure si n'hi ha molts

```
sum(is.na(DataValues_S013))/(dim(DataValues_S013)[1]*dim(DataValues_S013)[2])
```

```
## [1] 0.1248895
```

Això apunta a que tenim un 12,5% de valors desconeguts. Aquest és un aspecte que caldrà tenir en compte en els anàlisi posteriors

4. Generació de l'object SummarizedExperiment

Ara que ja sabem que tenim a les dades, podem procedir a carregar-les en un objecte *SummarizedExperiment*. Per a fer-ho, convertirem però les dades a un data frame i li traurem la primera columna que ja hem vist que és redundant amb la segona. Aleshores ja ho podem carregar. En concret, carregarem:

1. Les dades de les 9 primeres columnes (obviant la primera que està duplicada) de DataValues corresponen a dades dels pacients que els guardarem com a rowData ja que no formen part de l'anàlisi dels metabolits
2. La resta de dades de DataValues seràn els assays que és on es guarden les dades de les mostres que tenim
3. L'arxiu DataInfo com a colData que és on es guarda la informació de com és cada columna d'assays. D'aquí només hem d'agafar la part corresponent als metabolits. És a dir de la fila 11 endavant.

```
# Primer carreguem la llibreria per poder tractar el SummarizedExperiment
library(SummarizedExperiment)
```

```
## Loading required package: MatrixGenerics
```

```
## Loading required package: matrixStats
```

```
##
```

```
## Attaching package: 'MatrixGenerics'
```

```
## The following objects are masked from 'package:matrixStats':
```

```
##
```

```
## colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
## colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
## colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
## colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
## colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
## colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
## colWeightedMeans, colWeightedMedians, colWeightedSds,
## colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
## rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
## rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
## rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
## rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
## rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
## rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
## rowWeightedSds, rowWeightedVars
```

```
## Loading required package: GenomicRanges
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
##
```

```
## Attaching package: 'BiocGenerics'
```

```

## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##   match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##   Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##   table, tapply, union, unique, unsplit, which.max, which.min
## Loading required package: S4Vectors
##
## Attaching package: 'S4Vectors'
## The following object is masked from 'package:utils':
##
##   findMatches
## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname
## Loading required package: IRanges
##
## Attaching package: 'IRanges'
## The following object is masked from 'package:grDevices':
##
##   windows
## Loading required package: GenomeInfoDb
## Loading required package: Biobase
## Welcome to Bioconductor
##
##   Vignettes contain introductory material; view with
##   'browseVignettes()'. To cite Bioconductor, see
##   'citation("Biobase")', and for packages 'citation("pkgname)".
##
## Attaching package: 'Biobase'
## The following object is masked from 'package:MatrixGenerics':
##
##   rowMedians
## The following objects are masked from 'package:matrixStats':
##
##   anyMissing, rowMedians
# Després generem les dades de pacients i de metabolits i la llista de rownames que volem tal i com hem
pacients <- DataValues_S013[,2:10]
metabolits <- DataValues_S013[,11:696]
colNames <- DataInfo_S013[10:695,]
# Ara ja podem carregar el SummarizedExperiment

```

```
seMA <- SummarizedExperiment(assays = list(counts = as.matrix(metabolits)),
                             colData = colNames,
                             rowData = patients)

seMA
```

```
## class: SummarizedExperiment
## dim: 39 686
## metadata(0):
## assays(1): counts
## rownames: NULL
## rowData names(9): SUBJECTS SURGERY ... MEDINF_TO MEDHTA_TO
## colnames(686): GLU_TO INS_TO ... SM.C24.0_T5 SM.C24.1_T5
## colData names(4): X VarName varTpe Description
```

Sembla que l'objecte s'ha generat correctament

Mirem un parell de funcions que es poden fer.

```
# Comprovem com es diuen les columnes que tenim en cada una de les classes de dins de l'objecte
dimnames(seMA)
```

```
## [[1]]
## NULL
##
## [[2]]
##      [1] "GLU_TO"          "INS_TO"          "HOMA_TO"
##      [4] "HBA1C_TO"        "HBA1C.mmol.mol_TO" "PESO_TO"
##      [7] "bmi_TO"          "CC_TO"           "CINT_TO"
##     [10] "CAD_TO"          "TAD_TO"          "TAS_TO"
##     [13] "TG_TO"           "COL_TO"          "LDL_TO"
##     [16] "HDL_TO"          "VLDL_TO"         "PCR_TO"
##     [19] "LEP_TO"          "ADIPO_TO"        "GOT_TO"
##     [22] "GPT_TO"          "GGT_TO"          "URICO_TO"
##     [25] "CREAT_TO"        "UREA_TO"         "HIERRO_TO"
##     [28] "TRANSF_TO"       "FERR_TO"         "Ile_TO"
##     [31] "Leu_TO"          "Val_TO"          "Ala_TO"
##     [34] "Pro_TO"          "Gly_TO"          "Ser_TO"
##     [37] "Trp_TO"          "Phe_TO"          "Met_TO"
##     [40] "Orn_TO"          "Arg_TO"          "His_TO"
##     [43] "Asn_TO"          "Asp_TO"          "Glu_TO"
##     [46] "Gln_TO"          "Cit_TO"          "Tyr_TO"
##     [49] "Thr_TO"          "Lys_TO"          "Creatinine_TO"
##     [52] "Kynurenine_TO"   "Putrescine_TO"   "Sarcosine_TO"
##     [55] "Serotonin_TO"    "Taurine_TO"      "SDMA_TO"
##     [58] "CO_TO"           "C2_TO"           "C3.OH_TO"
##     [61] "C6..C4.1.DC._TO" "C5.DC..C6.OH._TO" "C7.DC_TO"
##     [64] "C8_TO"           "C10_TO"          "C10.1_TO"
##     [67] "C10.2_TO"        "C14.1_TO"        "C14.2_TO"
##     [70] "C16.1_TO"        "C16.2_TO"        "C16.2.OH_TO"
##     [73] "C18.1_TO"        "C18.1.OH_TO"     "C18.2_TO"
##     [76] "lysoPC.a.C16.0_TO" "lysoPC.a.C16.1_TO" "lysoPC.a.C17.0_TO"
##     [79] "lysoPC.a.C18.0_TO" "lysoPC.a.C18.1_TO" "lysoPC.a.C18.2_TO"
##     [82] "lysoPC.a.C20.3_TO" "lysoPC.a.C20.4_TO" "lysoPC.a.C24.0_TO"
##     [85] "lysoPC.a.C26.0_TO" "lysoPC.a.C26.1_TO" "lysoPC.a.C28.0_TO"
##     [88] "lysoPC.a.C28.1_TO" "PC.aa.C24.0_TO"   "PC.aa.C28.1_TO"
##     [91] "PC.aa.C30.0_TO"   "PC.aa.C32.0_TO"   "PC.aa.C32.1_TO"
```

## [94]	"PC.aa.C32.3_T0"	"PC.aa.C34.1_T0"	"PC.aa.C34.2_T0"
## [97]	"PC.aa.C34.3_T0"	"PC.aa.C34.4_T0"	"PC.aa.C36.0_T0"
## [100]	"PC.aa.C36.1_T0"	"PC.aa.C36.2_T0"	"PC.aa.C36.3_T0"
## [103]	"PC.aa.C36.4_T0"	"PC.aa.C36.5_T0"	"PC.aa.C38.0_T0"
## [106]	"PC.aa.C38.1_T0"	"PC.aa.C38.3_T0"	"PC.aa.C38.4_T0"
## [109]	"PC.aa.C38.5_T0"	"PC.aa.C38.6_T0"	"PC.aa.C40.1_T0"
## [112]	"PC.aa.C40.2_T0"	"PC.aa.C40.3_T0"	"PC.aa.C40.4_T0"
## [115]	"PC.aa.C40.5_T0"	"PC.aa.C40.6_T0"	"PC.aa.C42.0_T0"
## [118]	"PC.aa.C42.1_T0"	"PC.aa.C42.2_T0"	"PC.aa.C42.4_T0"
## [121]	"PC.aa.C42.5_T0"	"PC.aa.C42.6_T0"	"PC.aa.C30.0_T0"
## [124]	"PC.aa.C32.1_T0"	"PC.aa.C32.2_T0"	"PC.aa.C34.0_T0"
## [127]	"PC.aa.C34.1_T0"	"PC.aa.C34.2_T0"	"PC.aa.C34.3_T0"
## [130]	"PC.aa.C36.0_T0"	"PC.aa.C36.1_T0"	"PC.aa.C36.2_T0"
## [133]	"PC.aa.C36.3_T0"	"PC.aa.C36.4_T0"	"PC.aa.C36.5_T0"
## [136]	"PC.aa.C38.0_T0"	"PC.aa.C38.2_T0"	"PC.aa.C38.3_T0"
## [139]	"PC.aa.C38.4_T0"	"PC.aa.C38.5_T0"	"PC.aa.C38.6_T0"
## [142]	"PC.aa.C40.1_T0"	"PC.aa.C40.2_T0"	"PC.aa.C40.3_T0"
## [145]	"PC.aa.C40.4_T0"	"PC.aa.C40.5_T0"	"PC.aa.C40.6_T0"
## [148]	"PC.aa.C42.1_T0"	"PC.aa.C42.2_T0"	"PC.aa.C42.3_T0"
## [151]	"PC.aa.C42.4_T0"	"PC.aa.C42.5_T0"	"PC.aa.C44.3_T0"
## [154]	"PC.aa.C44.4_T0"	"PC.aa.C44.5_T0"	"PC.aa.C44.6_T0"
## [157]	"SM..OH..C14.1_T0"	"SM..OH..C16.1_T0"	"SM..OH..C22.1_T0"
## [160]	"SM..OH..C22.2_T0"	"SM..OH..C24.1_T0"	"SM.C16.0_T0"
## [163]	"SM.C16.1_T0"	"SM.C18.0_T0"	"SM.C18.1_T0"
## [166]	"SM.C20.2_T0"	"SM.C24.0_T0"	"SM.C24.1_T0"
## [169]	"MEDDM_T2"	"MEDCOL_T2"	"MEDINF_T2"
## [172]	"MEDHTA_T2"	"GLU_T2"	"INS_T2"
## [175]	"HOMA_T2"	"HBA1C_T2"	"HBA1C.mmol.mol_T2"
## [178]	"PESO_T2"	"bmi_T2"	"CC_T2"
## [181]	"CINT_T2"	"CAD_T2"	"TAD_T2"
## [184]	"TAS_T2"	"TG_T2"	"COL_T2"
## [187]	"LDL_T2"	"HDL_T2"	"VLDL_T2"
## [190]	"PCR_T2"	"LEP_T2"	"ADIPO_T2"
## [193]	"GOT_T2"	"GPT_T2"	"GGT_T2"
## [196]	"URICO_T2"	"CREAT_T2"	"UREA_T2"
## [199]	"HIERRO_T2"	"TRANSF_T2"	"FERR_T2"
## [202]	"Ile_T2"	"Leu_T2"	"Val_T2"
## [205]	"Ala_T2"	"Pro_T2"	"Gly_T2"
## [208]	"Ser_T2"	"Trp_T2"	"Phe_T2"
## [211]	"Met_T2"	"Orn_T2"	"Arg_T2"
## [214]	"His_T2"	"Asn_T2"	"Asp_T2"
## [217]	"Glu_T2"	"Gln_T2"	"Cit_T2"
## [220]	"Tyr_T2"	"Thr_T2"	"Lys_T2"
## [223]	"Creatinine_T2"	"Kynurenine_T2"	"Putrescine_T2"
## [226]	"Sarcosine_T2"	"Serotonin_T2"	"Taurine_T2"
## [229]	"SDMA_T2"	"C0_T2"	"C2_T2"
## [232]	"C3.OH_T2"	"C6..C4.1.DC._T2"	"C5.DC..C6.OH._T2"
## [235]	"C7.DC_T2"	"C8_T2"	"C10_T2"
## [238]	"C10.1_T2"	"C10.2_T2"	"C14.1_T2"
## [241]	"C14.2_T2"	"C16.1_T2"	"C16.2_T2"
## [244]	"C16.2.OH_T2"	"C18.1_T2"	"C18.1.OH_T2"
## [247]	"C18.2_T2"	"X"	"lysoPC.a.C14.0_T2"
## [250]	"lysoPC.a.C16.0_T2"	"lysoPC.a.C16.1_T2"	"lysoPC.a.C17.0_T2"
## [253]	"lysoPC.a.C18.0_T2"	"lysoPC.a.C18.1_T2"	"lysoPC.a.C18.2_T2"

## [256]	"lysoPC.a.C20.3_T2"	"lysoPC.a.C20.4_T2"	"lysoPC.a.C24.0_T2"
## [259]	"lysoPC.a.C26.0_T2"	"lysoPC.a.C26.1_T2"	"lysoPC.a.C28.0_T2"
## [262]	"lysoPC.a.C28.1_T2"	"PC.aa.C24.0_T2"	"PC.aa.C28.1_T2"
## [265]	"PC.aa.C30.0_T2"	"PC.aa.C32.0_T2"	"PC.aa.C32.1_T2"
## [268]	"PC.aa.C32.3_T2"	"PC.aa.C34.1_T2"	"PC.aa.C34.2_T2"
## [271]	"PC.aa.C34.3_T2"	"PC.aa.C34.4_T2"	"PC.aa.C36.0_T2"
## [274]	"PC.aa.C36.1_T2"	"PC.aa.C36.2_T2"	"PC.aa.C36.3_T2"
## [277]	"PC.aa.C36.4_T2"	"PC.aa.C36.5_T2"	"PC.aa.C38.0_T2"
## [280]	"PC.aa.C38.1_T2"	"PC.aa.C38.3_T2"	"PC.aa.C38.4_T2"
## [283]	"PC.aa.C38.5_T2"	"PC.aa.C38.6_T2"	"PC.aa.C40.1_T2"
## [286]	"PC.aa.C40.2_T2"	"PC.aa.C40.3_T2"	"PC.aa.C40.4_T2"
## [289]	"PC.aa.C40.5_T2"	"PC.aa.C40.6_T2"	"PC.aa.C42.0_T2"
## [292]	"PC.aa.C42.1_T2"	"PC.aa.C42.2_T2"	"PC.aa.C42.4_T2"
## [295]	"PC.aa.C42.5_T2"	"PC.aa.C42.6_T2"	"PC.ae.C30.0_T2"
## [298]	"PC.ae.C32.1_T2"	"PC.ae.C32.2_T2"	"PC.ae.C34.0_T2"
## [301]	"PC.ae.C34.1_T2"	"PC.ae.C34.2_T2"	"PC.ae.C34.3_T2"
## [304]	"PC.ae.C36.0_T2"	"PC.ae.C36.1_T2"	"PC.ae.C36.2_T2"
## [307]	"PC.ae.C36.3_T2"	"PC.ae.C36.4_T2"	"PC.ae.C36.5_T2"
## [310]	"PC.ae.C38.0_T2"	"PC.ae.C38.2_T2"	"PC.ae.C38.3_T2"
## [313]	"PC.ae.C38.4_T2"	"PC.ae.C38.5_T2"	"PC.ae.C38.6_T2"
## [316]	"PC.ae.C40.1_T2"	"PC.ae.C40.2_T2"	"PC.ae.C40.3_T2"
## [319]	"PC.ae.C40.4_T2"	"PC.ae.C40.5_T2"	"PC.ae.C40.6_T2"
## [322]	"PC.ae.C42.1_T2"	"PC.ae.C42.2_T2"	"PC.ae.C42.3_T2"
## [325]	"PC.ae.C42.4_T2"	"PC.ae.C42.5_T2"	"PC.ae.C44.3_T2"
## [328]	"PC.ae.C44.4_T2"	"PC.ae.C44.5_T2"	"PC.ae.C44.6_T2"
## [331]	"SM..OH..C14.1_T2"	"SM..OH..C16.1_T2"	"SM..OH..C22.1_T2"
## [334]	"SM..OH..C22.2_T2"	"SM..OH..C24.1_T2"	"SM.C16.0_T2"
## [337]	"SM.C16.1_T2"	"SM.C18.0_T2"	"SM.C18.1_T2"
## [340]	"SM.C20.2_T2"	"SM.C24.0_T2"	"SM.C24.1_T2"
## [343]	"MEDDM_T4"	"MEDCOL_T4"	"MEDINF_T4"
## [346]	"MEDHTA_T4"	"GLU_T4"	"INS_T4"
## [349]	"HOMA_T4"	"HBA1C_T4"	"HBA1C.mmol.mol_T4"
## [352]	"PESO_T4"	"bmi_T4"	"CC_T4"
## [355]	"CINT_T4"	"CAD_T4"	"TAD_T4"
## [358]	"TAS_T4"	"TG_T4"	"COL_T4"
## [361]	"LDL_T4"	"HDL_T4"	"VLDL_T4"
## [364]	"PCR_T4"	"LEP_T4"	"ADIPO_T4"
## [367]	"GOT_T4"	"GPT_T4"	"GGT_T4"
## [370]	"URICO_T4"	"CREAT_T4"	"UREA_T4"
## [373]	"HIERRO_T4"	"TRANSF_T4"	"FERR_T4"
## [376]	"Ile_T4"	"Leu_T4"	"Val_T4"
## [379]	"Ala_T4"	"Pro_T4"	"Gly_T4"
## [382]	"Ser_T4"	"Trp_T4"	"Phe_T4"
## [385]	"Met_T4"	"Orn_T4"	"Arg_T4"
## [388]	"His_T4"	"Asn_T4"	"Asp_T4"
## [391]	"Glu_T4"	"Gln_T4"	"Cit_T4"
## [394]	"Tyr_T4"	"Thr_T4"	"Lys_T4"
## [397]	"Creatinine_T4"	"Kynurenine_T4"	"Putrescine_T4"
## [400]	"Sarcosine_T4"	"Serotonin_T4"	"Taurine_T4"
## [403]	"SDMA_T4"	"CO_T4"	"C2_T4"
## [406]	"C3.OH_T4"	"C6..C4.1.DC.._T4"	"C5.DC..C6.OH.._T4"
## [409]	"C7.DC_T4"	"C8_T4"	"C10_T4"
## [412]	"C10.1_T4"	"C10.2_T4"	"C14.1_T4"
## [415]	"C14.2_T4"	"C16.1_T4"	"C16.2_T4"

## [418]	"C16.2.OH_T4"	"C18.1_T4"	"C18.1.OH_T4"
## [421]	"C18.2_T4"	"lysoPC.a.C16.0_T4"	"lysoPC.a.C16.1_T4"
## [424]	"lysoPC.a.C17.0_T4"	"lysoPC.a.C18.0_T4"	"lysoPC.a.C18.1_T4"
## [427]	"lysoPC.a.C18.2_T4"	"lysoPC.a.C20.3_T4"	"lysoPC.a.C20.4_T4"
## [430]	"lysoPC.a.C24.0_T4"	"lysoPC.a.C26.0_T4"	"lysoPC.a.C26.1_T4"
## [433]	"lysoPC.a.C28.0_T4"	"lysoPC.a.C28.1_T4"	"PC.aa.C24.0_T4"
## [436]	"PC.aa.C28.1_T4"	"PC.aa.C30.0_T4"	"PC.aa.C32.0_T4"
## [439]	"PC.aa.C32.1_T4"	"PC.aa.C32.3_T4"	"PC.aa.C34.1_T4"
## [442]	"PC.aa.C34.2_T4"	"PC.aa.C34.3_T4"	"PC.aa.C34.4_T4"
## [445]	"PC.aa.C36.0_T4"	"PC.aa.C36.1_T4"	"PC.aa.C36.2_T4"
## [448]	"PC.aa.C36.3_T4"	"PC.aa.C36.4_T4"	"PC.aa.C36.5_T4"
## [451]	"PC.aa.C38.0_T4"	"PC.aa.C38.1_T4"	"PC.aa.C38.3_T4"
## [454]	"PC.aa.C38.4_T4"	"PC.aa.C38.5_T4"	"PC.aa.C38.6_T4"
## [457]	"PC.aa.C40.1_T4"	"PC.aa.C40.2_T4"	"PC.aa.C40.3_T4"
## [460]	"PC.aa.C40.4_T4"	"PC.aa.C40.5_T4"	"PC.aa.C40.6_T4"
## [463]	"PC.aa.C42.0_T4"	"PC.aa.C42.1_T4"	"PC.aa.C42.2_T4"
## [466]	"PC.aa.C42.4_T4"	"PC.aa.C42.5_T4"	"PC.aa.C42.6_T4"
## [469]	"PC.aa.C30.0_T4"	"PC.aa.C32.1_T4"	"PC.aa.C32.2_T4"
## [472]	"PC.aa.C34.0_T4"	"PC.aa.C34.1_T4"	"PC.aa.C34.2_T4"
## [475]	"PC.aa.C34.3_T4"	"PC.aa.C36.0_T4"	"PC.aa.C36.1_T4"
## [478]	"PC.aa.C36.2_T4"	"PC.aa.C36.3_T4"	"PC.aa.C36.4_T4"
## [481]	"PC.aa.C36.5_T4"	"PC.aa.C38.0_T4"	"PC.aa.C38.2_T4"
## [484]	"PC.aa.C38.3_T4"	"PC.aa.C38.4_T4"	"PC.aa.C38.5_T4"
## [487]	"PC.aa.C38.6_T4"	"PC.aa.C40.1_T4"	"PC.aa.C40.2_T4"
## [490]	"PC.aa.C40.3_T4"	"PC.aa.C40.4_T4"	"PC.aa.C40.5_T4"
## [493]	"PC.aa.C40.6_T4"	"PC.aa.C42.1_T4"	"PC.aa.C42.2_T4"
## [496]	"PC.aa.C42.3_T4"	"PC.aa.C42.4_T4"	"PC.aa.C42.5_T4"
## [499]	"PC.aa.C44.3_T4"	"PC.aa.C44.4_T4"	"PC.aa.C44.5_T4"
## [502]	"PC.aa.C44.6_T4"	"SM..OH..C14.1_T4"	"SM..OH..C16.1_T4"
## [505]	"SM..OH..C22.1_T4"	"SM..OH..C22.2_T4"	"SM..OH..C24.1_T4"
## [508]	"SM.C16.0_T4"	"SM.C16.1_T4"	"SM.C18.0_T4"
## [511]	"SM.C18.1_T4"	"SM.C20.2_T4"	"SM.C24.0_T4"
## [514]	"SM.C24.1_T4"	"MEDDM_T5"	"MEDCOL_T5"
## [517]	"MEDINF_T5"	"MEDHTA_T5"	"GLU_T5"
## [520]	"INS_T5"	"HOMA_T5"	"HBA1C_T5"
## [523]	"HBA1C.mmol.mol_T5"	"PESO_T5"	"bmi_T5"
## [526]	"CC_T5"	"CINT_T5"	"CAD_T5"
## [529]	"TAD_T5"	"TAS_T5"	"TG_T5"
## [532]	"COL_T5"	"LDL_T5"	"HDL_T5"
## [535]	"VLDL_T5"	"PCR_T5"	"LEP_T5"
## [538]	"ADIPO_T5"	"GOT_T5"	"GPT_T5"
## [541]	"GGT_T5"	"URICO_T5"	"CREAT_T5"
## [544]	"UREA_T5"	"HIERRO_T5"	"TRANSF_T5"
## [547]	"FERR_T5"	"Ile_T5"	"Leu_T5"
## [550]	"Val_T5"	"Ala_T5"	"Pro_T5"
## [553]	"Gly_T5"	"Ser_T5"	"Trp_T5"
## [556]	"Phe_T5"	"Met_T5"	"Orn_T5"
## [559]	"Arg_T5"	"His_T5"	"Asn_T5"
## [562]	"Asp_T5"	"Glu_T5"	"Gln_T5"
## [565]	"Cit_T5"	"Tyr_T5"	"Thr_T5"
## [568]	"Lys_T5"	"Creatinine_T5"	"Kynurenine_T5"
## [571]	"Putrescine_T5"	"Sarcosine_T5"	"Serotonin_T5"
## [574]	"Taurine_T5"	"SDMA_T5"	"CO_T5"
## [577]	"C2_T5"	"C3.OH_T5"	"C6..C4.1.DC..T5"


```
## [580] "C5.DC..C6.OH._T5" "C7.DC_T5" "C8_T5"
## [583] "C10_T5" "C10.1_T5" "C10.2_T5"
## [586] "C14.1_T5" "C14.2_T5" "C16.1_T5"
## [589] "C16.2_T5" "C16.2.OH_T5" "C18.1_T5"
## [592] "C18.1.OH_T5" "C18.2_T5" "lysoPC.a.C16.0_T5"
## [595] "lysoPC.a.C16.1_T5" "lysoPC.a.C17.0_T5" "lysoPC.a.C18.0_T5"
## [598] "lysoPC.a.C18.1_T5" "lysoPC.a.C18.2_T5" "lysoPC.a.C20.3_T5"
## [601] "lysoPC.a.C20.4_T5" "lysoPC.a.C24.0_T5" "lysoPC.a.C26.0_T5"
## [604] "lysoPC.a.C26.1_T5" "lysoPC.a.C28.0_T5" "lysoPC.a.C28.1_T5"
## [607] "PC.aa.C24.0_T5" "PC.aa.C28.1_T5" "PC.aa.C30.0_T5"
## [610] "PC.aa.C32.0_T5" "PC.aa.C32.1_T5" "PC.aa.C32.3_T5"
## [613] "PC.aa.C34.1_T5" "PC.aa.C34.2_T5" "PC.aa.C34.3_T5"
## [616] "PC.aa.C34.4_T5" "PC.aa.C36.0_T5" "PC.aa.C36.1_T5"
## [619] "PC.aa.C36.2_T5" "PC.aa.C36.3_T5" "PC.aa.C36.4_T5"
## [622] "PC.aa.C36.5_T5" "PC.aa.C38.0_T5" "PC.aa.C38.1_T5"
## [625] "PC.aa.C38.3_T5" "PC.aa.C38.4_T5" "PC.aa.C38.5_T5"
## [628] "PC.aa.C38.6_T5" "PC.aa.C40.1_T5" "PC.aa.C40.2_T5"
## [631] "PC.aa.C40.3_T5" "PC.aa.C40.4_T5" "PC.aa.C40.5_T5"
## [634] "PC.aa.C40.6_T5" "PC.aa.C42.0_T5" "PC.aa.C42.1_T5"
## [637] "PC.aa.C42.2_T5" "PC.aa.C42.4_T5" "PC.aa.C42.5_T5"
## [640] "PC.aa.C42.6_T5" "PC.ae.C30.0_T5" "PC.ae.C32.1_T5"
## [643] "PC.ae.C32.2_T5" "PC.ae.C34.0_T5" "PC.ae.C34.1_T5"
## [646] "PC.ae.C34.2_T5" "PC.ae.C34.3_T5" "PC.ae.C36.0_T5"
## [649] "PC.ae.C36.1_T5" "PC.ae.C36.2_T5" "PC.ae.C36.3_T5"
## [652] "PC.ae.C36.4_T5" "PC.ae.C36.5_T5" "PC.ae.C38.0_T5"
## [655] "PC.ae.C38.2_T5" "PC.ae.C38.3_T5" "PC.ae.C38.4_T5"
## [658] "PC.ae.C38.5_T5" "PC.ae.C38.6_T5" "PC.ae.C40.1_T5"
## [661] "PC.ae.C40.2_T5" "PC.ae.C40.3_T5" "PC.ae.C40.4_T5"
## [664] "PC.ae.C40.5_T5" "PC.ae.C40.6_T5" "PC.ae.C42.1_T5"
## [667] "PC.ae.C42.2_T5" "PC.ae.C42.3_T5" "PC.ae.C42.4_T5"
## [670] "PC.ae.C42.5_T5" "PC.ae.C44.3_T5" "PC.ae.C44.4_T5"
## [673] "PC.ae.C44.5_T5" "PC.ae.C44.6_T5" "SM..OH..C14.1_T5"
## [676] "SM..OH..C16.1_T5" "SM..OH..C22.1_T5" "SM..OH..C22.2_T5"
## [679] "SM..OH..C24.1_T5" "SM.C16.0_T5" "SM.C16.1_T5"
## [682] "SM.C18.0_T5" "SM.C18.1_T5" "SM.C20.2_T5"
## [685] "SM.C24.0_T5" "SM.C24.1_T5"
```

```
# I ara mirem les 10 primeres columnes i les 5 primeres files d'assays
assay(seMA)[1:5, 1:10]
```

```
##      GLU_TO INS_TO HOMA_TO HBA1C_TO HBA1C.mmol.mol_TO PESO_TO bmi_TO CC_TO
## [1,]    85  11.40   2.40      NA              NA      151   62.9   0.7
## [2,]    78  12.10   2.32      NA              NA      139   47.0   NA
## [3,]    75   8.41   1.56    5.4          35.51      84   29.8   0.7
## [4,]    71  12.80   2.25    5.1          32.23     136   53.1   1.0
## [5,]    82   6.01   1.22    5.6          37.69     121   46.6   0.9
##      CINT_TO CAD_TO
## [1,]    116    167
## [2,]     NA     NA
## [3,]     90    126
## [4,]    157    162
## [5,]    123    132
```

```
# I també mirem com ha quedat colData
colData(seMA)
```

```
## DataFrame with 686 rows and 4 columns
##           X           VarName      varTpe Description
##           <character> <character> <character> <character>
## GLU_T0          GLU_T0          GLU_T0      integer    dataDesc
## INS_T0          INS_T0          INS_T0      numeric     dataDesc
## HOMA_T0         HOMA_T0         HOMA_T0      numeric     dataDesc
## HBA1C_T0        HBA1C_T0        HBA1C_T0      numeric     dataDesc
## HBA1C.mmol.mol_T0 HBA1C.mmol.mol_T0 HBA1C.mmol.mol_T0 numeric     dataDesc
## ...           ...           ...           ...           ...
## SM.C18.0_T5      SM.C18.0_T5      SM.C18.0_T5      numeric     dataDesc
## SM.C18.1_T5      SM.C18.1_T5      SM.C18.1_T5      numeric     dataDesc
## SM.C20.2_T5      SM.C20.2_T5      SM.C20.2_T5      numeric     dataDesc
## SM.C24.0_T5      SM.C24.0_T5      SM.C24.0_T5      numeric     dataDesc
## SM.C24.1_T5      SM.C24.1_T5      SM.C24.1_T5      numeric     dataDesc

# I les dades bàsiques dels pacients
rowData(seMA)
```

```
## DataFrame with 39 rows and 9 columns
##      SUBJECTS  SURGERY    AGE    GENDER    Group  MEDDM_T0  MEDCOL_T0
##      <integer> <character> <integer> <character> <integer> <integer> <integer>
## 1           1    by pass     27         F         1         0         0
## 2           2    by pass     19         F         2         0         0
## 3           3    by pass     42         F         1         0         0
## 4           4    by pass     37         F         2         0         0
## 5           5    tubular     42         F         1         0         0
## ...         ...         ...         ...         ...         ...         ...
## 35          35    tubular     39         M         2         0         0
## 36          36    tubular     35         M         1         0         0
## 37          37    by pass     46         M         2         0         0
## 38          38    tubular     41         M         1         0         0
## 39          39    by pass     26         M         1         0         0
##      MEDINF_T0  MEDHTA_T0
##      <integer> <integer>
## 1           0         1
## 2           0         0
## 3           0         0
## 4           0         0
## 5           0         0
## ...         ...         ...
## 35          0         1
## 36          0         0
## 37          1         0
## 38          0         0
## 39          0         0
```

5. Anàlisi exploratoria inicial de les dades

5.1. Descripció de les dades

Fins ara hem vist que les nostres dades tenen NA, el que pot suposar un problema. Anem a veure també si la variabilitat de rangs és molt gran i ens cal normalitzar les dades. Fem un summary a més per veure com varien els estadístics bàsics. Si ho fem amb tot l'arxiu no s'arriba a apreciar la sortida, així que en fem uns quants

```
summary(assay(seMA)[,1:150])
```

```
##      GLU_TO      INS_TO      HOMA_TO      HBA1C_TO
## Min.   : 71.0   Min.   : 3.40   Min.   : 0.760   Min.   :5.100
## 1st Qu.: 91.0   1st Qu.:11.85   1st Qu.: 2.435   1st Qu.:5.400
## Median :103.0   Median :16.00   Median : 4.210   Median :5.600
## Mean   :106.5   Mean   :17.60   Mean   : 4.890   Mean   :5.592
## 3rd Qu.:109.5   3rd Qu.:21.15   3rd Qu.: 5.720   3rd Qu.:5.800
## Max.   :263.0   Max.   :43.00   Max.   :13.600   Max.   :6.400
##                                     NA's   :15
## HBA1C.mmol.mol_TO  PESO_TO      bmi_TO      CC_TO
## Min.   :32.23   Min.   : 84.0   Min.   :29.80   Min.   :0.7000
## 1st Qu.:35.51   1st Qu.:119.5   1st Qu.:44.40   1st Qu.:0.9000
## Median :37.69   Median :135.0   Median :48.80   Median :0.9000
## Mean   :37.60   Mean   :140.0   Mean   :50.52   Mean   :0.9343
## 3rd Qu.:39.88   3rd Qu.:155.0   3rd Qu.:55.35   3rd Qu.:1.0000
## Max.   :46.44   Max.   :200.0   Max.   :68.60   Max.   :1.7000
## NA's   :15                                     NA's   :4
##      CINT_TO      CAD_TO      TAD_TO      TAS_TO
## Min.   : 90.0   Min.   : 85.0   Min.   : 60.00   Min.   :111.0
## 1st Qu.:122.5   1st Qu.:136.0   1st Qu.: 74.50   1st Qu.:121.0
## Median :133.0   Median :148.0   Median : 83.00   Median :128.0
## Mean   :135.7   Mean   :147.1   Mean   : 83.58   Mean   :133.5
## 3rd Qu.:147.5   3rd Qu.:158.5   3rd Qu.: 90.00   3rd Qu.:144.0
## Max.   :186.0   Max.   :182.0   Max.   :125.00   Max.   :180.0
## NA's   :4      NA's   :4      NA's   :8      NA's   :8
##      TG_TO      COL_TO      LDL_TO      HDL_TO
## Min.   : 30.0   Min.   :102.0   Min.   : 24.0   Min.   : 11.00
## 1st Qu.: 84.0   1st Qu.:180.5   1st Qu.:100.3   1st Qu.: 39.25
## Median :118.0   Median :207.0   Median :130.0   Median : 47.00
## Mean   :129.1   Mean   :206.6   Mean   :128.2   Mean   : 49.97
## 3rd Qu.:151.0   3rd Qu.:236.0   3rd Qu.:158.8   3rd Qu.: 59.50
## Max.   :268.0   Max.   :295.0   Max.   :191.0   Max.   :107.00
##                                     NA's   :1   NA's   :1
##      VLDL_TO      PCR_TO      LEP_TO      ADIPO_TO
## Min.   : 6.00   Min.   : 0.00   Min.   : 27.00   Min.   : 1.720
## 1st Qu.:16.80   1st Qu.: 3.51   1st Qu.: 49.25   1st Qu.: 4.700
## Median :23.60   Median : 8.25   Median : 67.20   Median : 7.460
## Mean   :25.81   Mean   :10.13   Mean   : 78.33   Mean   : 7.702
## 3rd Qu.:30.20   3rd Qu.:13.90   3rd Qu.: 98.25   3rd Qu.: 8.590
## Max.   :53.60   Max.   :29.80   Max.   :155.00   Max.   :17.100
##                                     NA's   :14   NA's   :14
##      GOT_TO      GPT_TO      GGT_TO      URICO_TO
## Min.   :11.0   Min.   : 11.00   Min.   :12.00   Min.   :2.700
## 1st Qu.:15.0   1st Qu.: 31.00   1st Qu.:20.00   1st Qu.:4.700
## Median :20.0   Median : 40.00   Median :26.00   Median :5.500
## Mean   :23.9   Mean   : 44.54   Mean   :31.41   Mean   :5.501
## 3rd Qu.:25.0   3rd Qu.: 48.50   3rd Qu.:37.00   3rd Qu.:6.200
## Max.   :83.0   Max.   :157.00   Max.   :86.00   Max.   :8.600
##                                     NA's   :2   NA's   :2
##      CREAT_TO      UREA_TO      HIERRO_TO      TRANSF_TO
## Min.   :0.5000   Min.   :14.00   Min.   : 13.00   Min.   :159.0
## 1st Qu.:0.7000   1st Qu.:24.00   1st Qu.: 50.00   1st Qu.:236.5
## Median :0.8000   Median :29.00   Median : 69.00   Median :268.5
```

```

## Mean :0.7923 Mean :31.38 Mean : 75.61 Mean :262.9
## 3rd Qu.:0.9000 3rd Qu.:38.00 3rd Qu.: 92.25 3rd Qu.:289.8
## Max. :1.2000 Max. :56.00 Max. :166.00 Max. :374.0
## NA's :1 NA's :9
## FERR_T0 Ile_T0 Leu_T0 Val_T0
## Min. : 11.10 Min. : 53.50 Min. : 90.1 Min. :183.0
## 1st Qu.: 25.00 1st Qu.: 83.55 1st Qu.:150.5 1st Qu.:256.0
## Median : 49.00 Median : 91.70 Median :183.0 Median :311.0
## Mean : 77.06 Mean : 98.98 Mean :185.2 Mean :318.4
## 3rd Qu.: 97.00 3rd Qu.:116.00 3rd Qu.:217.0 3rd Qu.:369.5
## Max. :372.00 Max. :150.00 Max. :281.0 Max. :508.0
## NA's :2
## Ala_T0 Pro_T0 Gly_T0 Ser_T0
## Min. :194.0 Min. :123.0 Min. :153.0 Min. : 94.5
## 1st Qu.:414.0 1st Qu.:164.5 1st Qu.:227.5 1st Qu.:133.5
## Median :459.0 Median :222.0 Median :270.0 Median :159.0
## Mean :471.9 Mean :234.6 Mean :291.1 Mean :161.4
## 3rd Qu.:543.0 3rd Qu.:257.5 3rd Qu.:311.5 3rd Qu.:180.0
## Max. :907.0 Max. :592.0 Max. :586.0 Max. :310.0
## Trp_T0 Phe_T0 Met_T0 Orn_T0
## Min. : 41.00 Min. : 48.40 Min. : -99.00 Min. : 39.6
## 1st Qu.: 52.15 1st Qu.: 70.60 1st Qu.: 23.25 1st Qu.: 74.6
## Median : 62.30 Median : 87.30 Median : 28.30 Median :103.0
## Mean : 67.22 Mean : 87.48 Mean : 23.08 Mean :102.4
## 3rd Qu.: 76.95 3rd Qu.:102.00 3rd Qu.: 33.30 3rd Qu.:128.5
## Max. :123.00 Max. :139.00 Max. : 53.00 Max. :195.0
## Arg_T0 His_T0 Asn_T0 Asp_T0
## Min. : 90.7 Min. : 70.50 Min. :27.30 Min. : 7.10
## 1st Qu.:120.5 1st Qu.: 80.65 1st Qu.:39.55 1st Qu.:15.65
## Median :143.0 Median : 94.40 Median :47.40 Median :20.50
## Mean :148.5 Mean : 97.68 Mean :48.81 Mean :23.12
## 3rd Qu.:167.5 3rd Qu.:108.00 3rd Qu.:56.00 3rd Qu.:27.85
## Max. :254.0 Max. :167.00 Max. :82.10 Max. :59.30
## Glu_T0 Gln_T0 Cit_T0 Tyr_T0
## Min. : 16.90 Min. : 241.0 Min. :12.00 Min. : 49.30
## 1st Qu.: 51.70 1st Qu.: 628.0 1st Qu.:26.25 1st Qu.: 71.65
## Median : 66.40 Median : 713.0 Median :30.30 Median : 89.00
## Mean : 84.68 Mean : 749.9 Mean :32.76 Mean : 92.46
## 3rd Qu.: 86.25 3rd Qu.: 872.5 3rd Qu.:38.35 3rd Qu.:108.00
## Max. :351.00 Max. :1320.0 Max. :59.50 Max. :162.00
## Thr_T0 Lys_T0 Creatinine_T0 Kynurenine_T0
## Min. : 67.5 Min. :266.0 Min. : 39.10 Min. :1.610
## 1st Qu.:112.0 1st Qu.:316.5 1st Qu.: 65.85 1st Qu.:2.540
## Median :144.0 Median :378.0 Median : 78.40 Median :3.160
## Mean :151.5 Mean :381.8 Mean : 80.43 Mean :3.303
## 3rd Qu.:177.5 3rd Qu.:434.5 3rd Qu.: 99.20 3rd Qu.:3.775
## Max. :286.0 Max. :562.0 Max. :115.00 Max. :6.510
## Putrescine_T0 Sarcosine_T0 Serotonin_T0 Taurine_T0
## Min. : -99.0000 Min. : 2.050 Min. : -99.0000 Min. : 41.00

```

```

## 1st Qu.: 0.1295    1st Qu.: 5.895    1st Qu.: 0.3565    1st Qu.: 78.55
## Median : 0.2120    Median : 7.700    Median : 0.6500    Median :107.00
## Mean   :-12.7355    Mean   : 8.648    Mean   : -9.5029    Mean   :108.77
## 3rd Qu.: 0.2670    3rd Qu.:12.200    3rd Qu.: 0.9180    3rd Qu.:132.50
## Max.    : 0.3660    Max.    :15.700    Max.    : 1.8100    Max.    :181.00
##
##      SDMA_T0      C0_T0      C2_T0      C3.OH_T0
## Min.   :0.820    Min.   :21.60    Min.   : 0.360    Min.   :0.1500
## 1st Qu.:1.115    1st Qu.:36.00    1st Qu.: 5.215    1st Qu.:0.2100
## Median :1.290    Median :43.30    Median : 7.270    Median :0.2400
## Mean   :1.317    Mean   :45.60    Mean   : 7.762    Mean   :0.2474
## 3rd Qu.:1.535    3rd Qu.:52.05    3rd Qu.: 9.200    3rd Qu.:0.2800
## Max.   :1.940    Max.   :79.30    Max.   :19.500    Max.   :0.3700
##
## C6..C4.1.DC._T0  C5.DC..C6.OH._T0  C7.DC_T0  C8_T0
## Min.   :-99.000    Min.   :-9.0000    Min.   :-9.0000    Min.   :0.3230
## 1st Qu.: -9.000    1st Qu.: 0.0410    1st Qu.: 0.0395    1st Qu.:0.4385
## Median : 0.094    Median : 0.0490    Median : 0.0470    Median :0.5190
## Mean   :-14.232    Mean   :-1.1096    Mean   :-1.1098    Mean   :0.5419
## 3rd Qu.: 0.118    3rd Qu.: 0.0565    3rd Qu.: 0.0555    3rd Qu.:0.6235
## Max.    : 0.201    Max.    : 0.0780    Max.    : 0.0760    Max.    :0.8710
##
##      C10_T0      C10.1_T0      C10.2_T0      C14.1_T0
## Min.   :0.2500    Min.   :0.1300    Min.   :0.1000    Min.   :0.0900
## 1st Qu.:0.4200    1st Qu.:0.1900    1st Qu.:0.1300    1st Qu.:0.1450
## Median :0.5200    Median :0.2300    Median :0.1500    Median :0.1900
## Mean   :0.5274    Mean   :0.2482    Mean   :0.1621    Mean   :0.1944
## 3rd Qu.:0.6350    3rd Qu.:0.3050    3rd Qu.:0.1850    3rd Qu.:0.2450
## Max.   :0.9300    Max.   :0.4200    Max.   :0.2600    Max.   :0.3000
##
##      C14.2_T0      C16.1_T0      C16.2_T0      C16.2.OH_T0
## Min.   :0.0600    Min.   :0.0600    Min.   :0.03000    Min.   :0.00000
## 1st Qu.:0.1000    1st Qu.:0.0900    1st Qu.:0.03000    1st Qu.:0.04000
## Median :0.1200    Median :0.1000    Median :0.04000    Median :0.04000
## Mean   :0.1244    Mean   :0.1126    Mean   :0.04487    Mean   :0.04385
## 3rd Qu.:0.1400    3rd Qu.:0.1300    3rd Qu.:0.05000    3rd Qu.:0.05500
## Max.   :0.2100    Max.   :0.1900    Max.   :0.08000    Max.   :0.07000
##
##      C18.1_T0      C18.1.OH_T0      C18.2_T0      lysoPC.a.C16.0_T0
## Min.   :0.0700    Min.   :0.03000    Min.   :0.0600    Min.   : 72.3
## 1st Qu.:0.1200    1st Qu.:0.04000    1st Qu.:0.0900    1st Qu.:118.5
## Median :0.1600    Median :0.04000    Median :0.1100    Median :144.0
## Mean   :0.1703    Mean   :0.04615    Mean   :0.1074    Mean   :151.7
## 3rd Qu.:0.2150    3rd Qu.:0.05000    3rd Qu.:0.1300    3rd Qu.:174.0
## Max.   :0.2900    Max.   :0.07000    Max.   :0.1700    Max.   :308.0
##
## lysoPC.a.C16.1_T0 lysoPC.a.C17.0_T0 lysoPC.a.C18.0_T0 lysoPC.a.C18.1_T0
## Min.   :1.940    Min.   :0.790    Min.   : 14.30    Min.   :12.70
## 1st Qu.:2.870    1st Qu.:1.445    1st Qu.: 34.05    1st Qu.:22.25
## Median :3.810    Median :2.110    Median : 45.20    Median :28.80
## Mean   :4.283    Mean   :2.101    Mean   : 50.51    Mean   :32.32
## 3rd Qu.:5.500    3rd Qu.:2.425    3rd Qu.: 60.70    3rd Qu.:40.55
## Max.   :9.890    Max.   :4.340    Max.   :120.00    Max.   :58.20
##

```

```

## lysoPC.a.C18.2_T0 lysoPC.a.C20.3_T0 lysoPC.a.C20.4_T0 lysoPC.a.C24.0_T0
## Min. : 19.10 Min. :1.190 Min. : 4.21 Min. :0.2400
## 1st Qu.: 34.75 1st Qu.:2.940 1st Qu.: 7.50 1st Qu.:0.3500
## Median : 39.90 Median :4.090 Median :10.50 Median :0.4600
## Mean : 43.82 Mean :4.373 Mean :10.77 Mean :0.4974
## 3rd Qu.: 50.75 3rd Qu.:5.335 3rd Qu.:13.55 3rd Qu.:0.5550
## Max. :109.00 Max. :9.410 Max. :19.60 Max. :0.9200
##
## lysoPC.a.C26.0_T0 lysoPC.a.C26.1_T0 lysoPC.a.C28.0_T0 lysoPC.a.C28.1_T0
## Min. : -9.0000 Min. :0.1200 Min. : -9.0000 Min. :0.2400
## 1st Qu.: 0.3100 1st Qu.:0.2600 1st Qu.: 0.2650 1st Qu.:0.3800
## Median : 0.4200 Median :0.3300 Median : 0.3400 Median :0.4600
## Mean : 0.2841 Mean :0.3408 Mean : -0.7536 Mean :0.5238
## 3rd Qu.: 0.5050 3rd Qu.:0.4150 3rd Qu.: 0.4100 3rd Qu.:0.5900
## Max. : 2.1300 Max. :0.6600 Max. : 1.7200 Max. :1.5000
##
## PC.aa.C24.0_T0 PC.aa.C28.1_T0 PC.aa.C30.0_T0 PC.aa.C32.0_T0
## Min. : -9.000 Min. :1.970 Min. :0.630 Min. : 5.360
## 1st Qu.: 0.145 1st Qu.:2.380 1st Qu.:1.420 1st Qu.: 8.525
## Median : 0.180 Median :2.910 Median :1.710 Median :10.200
## Mean : -1.650 Mean :3.014 Mean :1.895 Mean :10.695
## 3rd Qu.: 0.250 3rd Qu.:3.640 3rd Qu.:2.305 3rd Qu.:12.100
## Max. : 0.780 Max. :4.770 Max. :3.520 Max. :17.300
##
## PC.aa.C32.1_T0 PC.aa.C32.3_T0 PC.aa.C34.1_T0 PC.aa.C34.2_T0
## Min. : 2.64 Min. :0.1200 Min. : 50.60 Min. :154.0
## 1st Qu.: 7.11 1st Qu.:0.1750 1st Qu.: 99.75 1st Qu.:245.0
## Median :10.50 Median :0.2400 Median :136.00 Median :302.0
## Mean :11.77 Mean :0.2362 Mean :139.15 Mean :338.7
## 3rd Qu.:12.80 3rd Qu.:0.2800 3rd Qu.:161.00 3rd Qu.:389.5
## Max. :29.60 Max. :0.4400 Max. :310.00 Max. :904.0
##
## PC.aa.C34.3_T0 PC.aa.C34.4_T0 PC.aa.C36.0_T0 PC.aa.C36.1_T0
## Min. : 3.94 Min. :0.2000 Min. :0.610 Min. : 8.40
## 1st Qu.: 6.34 1st Qu.:0.3700 1st Qu.:1.030 1st Qu.:13.30
## Median : 9.73 Median :0.6300 Median :1.300 Median :15.70
## Mean :10.29 Mean :0.7015 Mean :1.568 Mean :18.32
## 3rd Qu.:11.00 3rd Qu.:0.9050 3rd Qu.:1.880 3rd Qu.:23.00
## Max. :24.40 Max. :1.8600 Max. :4.560 Max. :43.30
##
## PC.aa.C36.2_T0 PC.aa.C36.3_T0 PC.aa.C36.4_T0 PC.aa.C36.5_T0
## Min. : 60.6 Min. : 36.60 Min. : 40.10 Min. : 2.770
## 1st Qu.: 89.8 1st Qu.: 58.90 1st Qu.: 95.15 1st Qu.: 6.595
## Median :112.0 Median : 69.80 Median :120.00 Median : 9.200
## Mean :125.9 Mean : 78.45 Mean :120.26 Mean : 9.591
## 3rd Qu.:151.0 3rd Qu.: 91.20 3rd Qu.:149.50 3rd Qu.:11.750
## Max. :282.0 Max. :187.00 Max. :214.00 Max. :23.700
##
## PC.aa.C38.0_T0 PC.aa.C38.1_T0 PC.aa.C38.3_T0 PC.aa.C38.4_T0
## Min. :1.340 Min. : -9.000 Min. : 16.90 Min. : 34.20
## 1st Qu.:2.385 1st Qu.: 0.910 1st Qu.: 29.15 1st Qu.: 66.85
## Median :3.000 Median : 1.100 Median : 37.40 Median : 92.20
## Mean :3.258 Mean : 1.219 Mean : 41.54 Mean : 92.36
## 3rd Qu.:3.960 3rd Qu.: 1.625 3rd Qu.: 51.25 3rd Qu.:111.50

```

```

## Max. :6.180 Max. : 6.230 Max. :110.00 Max. :164.00
##
## PC.aa.C38.5_T0 PC.aa.C38.6_T0 PC.aa.C40.1_T0 PC.aa.C40.2_T0
## Min. :11.20 Min. : 13.90 Min. :-9.000 Min. :0.1700
## 1st Qu.:20.55 1st Qu.: 43.90 1st Qu.: 0.335 1st Qu.:0.2300
## Median :29.50 Median : 56.70 Median : 0.460 Median :0.3400
## Mean :28.74 Mean : 62.95 Mean :-1.550 Mean :0.6377
## 3rd Qu.:36.75 3rd Qu.: 77.45 3rd Qu.: 0.565 3rd Qu.:0.4300
## Max. :47.70 Max. :128.00 Max. : 3.330 Max. :5.7700
##
## PC.aa.C40.3_T0 PC.aa.C40.4_T0 PC.aa.C40.5_T0 PC.aa.C40.6_T0
## Min. :0.370 Min. :1.080 Min. : 1.750 Min. : 5.39
## 1st Qu.:0.490 1st Qu.:1.860 1st Qu.: 3.680 1st Qu.:16.15
## Median :0.580 Median :2.500 Median : 4.910 Median :19.50
## Mean :0.871 Mean :2.777 Mean : 5.178 Mean :22.21
## 3rd Qu.:0.780 3rd Qu.:3.385 3rd Qu.: 6.185 3rd Qu.:29.65
## Max. :4.820 Max. :5.970 Max. :11.100 Max. :41.10
##
## PC.aa.C42.0_T0 PC.aa.C42.1_T0 PC.aa.C42.2_T0 PC.aa.C42.4_T0
## Min. :0.3700 Min. :0.2000 Min. :0.1600 Min. :0.1100
## 1st Qu.:0.5600 1st Qu.:0.2750 1st Qu.:0.2150 1st Qu.:0.1800
## Median :0.6800 Median :0.3500 Median :0.2800 Median :0.2200
## Mean :0.7562 Mean :0.4087 Mean :0.4064 Mean :0.3379
## 3rd Qu.:0.8700 3rd Qu.:0.4250 3rd Qu.:0.3300 3rd Qu.:0.2450
## Max. :1.7400 Max. :1.3800 Max. :2.3100 Max. :2.6100
##
## PC.aa.C42.5_T0 PC.aa.C42.6_T0 PC.aa.C30.0_T0 PC.aa.C32.1_T0
## Min. :0.1300 Min. :-9.00000 Min. :0.2100 Min. :1.220
## 1st Qu.:0.2600 1st Qu.: 0.27500 1st Qu.:0.2700 1st Qu.:1.675
## Median :0.3200 Median : 0.34000 Median :0.3300 Median :2.160
## Mean :0.3818 Mean :-0.08128 Mean :0.3262 Mean :2.219
## 3rd Qu.:0.4000 3rd Qu.: 0.42500 3rd Qu.:0.3600 3rd Qu.:2.625
## Max. :1.4400 Max. : 1.09000 Max. :0.5000 Max. :4.120
##
## PC.aa.C32.2_T0 PC.aa.C34.0_T0 PC.aa.C34.1_T0 PC.aa.C34.2_T0
## Min. :0.3500 Min. :0.250 Min. :2.760 Min. : 3.030
## 1st Qu.:0.5450 1st Qu.:0.390 1st Qu.:4.135 1st Qu.: 5.310
## Median :0.6700 Median :0.470 Median :4.850 Median : 6.170
## Mean :0.6997 Mean :0.511 Mean :5.147 Mean : 7.382
## 3rd Qu.:0.8300 3rd Qu.:0.630 3rd Qu.:6.395 3rd Qu.: 8.590
## Max. :1.2000 Max. :0.840 Max. :9.350 Max. :19.000
##
## PC.aa.C34.3_T0 PC.aa.C36.0_T0 PC.aa.C36.1_T0 PC.aa.C36.2_T0
## Min. : 1.450 Min. :0.470 Min. : 1.510 Min. : 3.230
## 1st Qu.: 3.285 1st Qu.:0.760 1st Qu.: 2.310 1st Qu.: 4.390
## Median : 4.140 Median :0.940 Median : 3.120 Median : 5.210
## Mean : 4.581 Mean :1.008 Mean : 3.922 Mean : 5.852
## 3rd Qu.: 5.030 3rd Qu.:1.135 3rd Qu.: 3.695 3rd Qu.: 6.965
## Max. :10.100 Max. :1.980 Max. :18.200 Max. :10.800
##
## PC.aa.C36.3_T0 PC.aa.C36.4_T0 PC.aa.C36.5_T0 PC.aa.C38.0_T0
## Min. : 1.660 Min. : 2.180 Min. : 1.760 Min. :0.490
## 1st Qu.: 2.710 1st Qu.: 6.490 1st Qu.: 4.950 1st Qu.:1.265
## Median : 3.120 Median : 7.940 Median : 6.240 Median :1.630

```

```
## Mean : 3.812 Mean : 8.826 Mean : 6.642 Mean :1.623
## 3rd Qu.: 4.610 3rd Qu.:10.850 3rd Qu.: 8.215 3rd Qu.:1.945
## Max. :11.000 Max. :20.100 Max. :13.100 Max. :2.960
##
## PC.ae.C38.2_T0 PC.ae.C38.3_T0 PC.ae.C38.4_T0 PC.ae.C38.5_T0
## Min. : 0.350 Min. : 0.830 Min. : 3.100 Min. : 3.370
## 1st Qu.: 1.195 1st Qu.: 1.250 1st Qu.: 4.440 1st Qu.: 7.470
## Median : 1.580 Median : 1.530 Median : 5.480 Median : 8.610
## Mean : 2.752 Mean : 2.235 Mean : 5.729 Mean : 9.477
## 3rd Qu.: 2.115 3rd Qu.: 1.990 3rd Qu.: 6.425 3rd Qu.:10.500
## Max. :25.500 Max. :11.200 Max. :11.100 Max. :19.200
##
## PC.ae.C38.6_T0 PC.ae.C40.1_T0 PC.ae.C40.2_T0 PC.ae.C40.3_T0
## Min. :1.380 Min. :0.450 Min. :0.620 Min. : 0.560
## 1st Qu.:2.715 1st Qu.:0.895 1st Qu.:0.965 1st Qu.: 0.740
## Median :3.740 Median :1.160 Median :1.170 Median : 0.880
## Mean :3.995 Mean :1.373 Mean :1.557 Mean : 1.574
## 3rd Qu.:4.915 3rd Qu.:1.570 3rd Qu.:1.390 3rd Qu.: 1.160
## Max. :7.690 Max. :4.540 Max. :8.440 Max. :11.400
##
## PC.ae.C40.4_T0 PC.ae.C40.5_T0 PC.ae.C40.6_T0 PC.ae.C42.1_T0
## Min. :1.020 Min. :1.050 Min. :1.340 Min. :0.2100
## 1st Qu.:1.645 1st Qu.:1.570 1st Qu.:1.700 1st Qu.:0.3600
## Median :1.940 Median :1.790 Median :2.200 Median :0.4300
## Mean :2.380 Mean :2.317 Mean :2.309 Mean :0.6195
## 3rd Qu.:2.610 3rd Qu.:2.385 3rd Qu.:3.035 3rd Qu.:0.5650
## Max. :7.970 Max. :8.690 Max. :3.540 Max. :3.4000
##
## PC.ae.C42.2_T0 PC.ae.C42.3_T0
## Min. :0.2900 Min. :0.4600
## 1st Qu.:0.3800 1st Qu.:0.6050
## Median :0.4700 Median :0.7900
## Mean :0.6236 Mean :0.9569
## 3rd Qu.:0.6100 3rd Qu.:0.9450
## Max. :2.8200 Max. :4.1900
##
```

Veiem que la variabilitat en els rangs és molt elevada i, per tant, caldrà normalitzar les dades. Recordem també que tenim un nombre de NAs força elevat.

5.2. Preprocessat

Amb el que hem pogut veure fins aquí de les dades que tenim és que ens calen dos processos per poder començar els anàlisi multivariants:

1. Imputar els missing values
2. Normalitzar les dades

Això ho podem fer amb POMA

```
# primer instal·lem el paquet i les llibreries
# BiocManager::install("POMA") # Això només el primer cop per instal·lar el paquet
library(POMA)
```

```
## Welcome to POMA!
## Version 1.16.0
```



```
## POMAShiny app: https://github.com/pcastellanoescuder/POMAShiny
```

```
library(ggtext)
```

```
## Warning: package 'ggtext' was built under R version 4.4.3
```

```
library(magrittr)
```

```
##
```

```
## Attaching package: 'magrittr'
```

```
## The following object is masked from 'package:GenomicRanges':
```

```
##
```

```
## subtract
```

Ataquem primer la imputació.

Abans de fer-la cal dir que no podem imputar NA en columnes on no tenim cap valor. Per tant, mirem si hi ha alguna columna que no tingui cap valor i l'eliminem.

```
# Busquem columnes amb tot NAs
```

```
eliminats = c()
```

```
for (i in 1:685)
```

```
  if (sum(is.na(assay(seMA)[,i])) == 39)
```

```
  {print((paste(i, sum(is.na(assay(seMA)[,i])))))
```

```
    eliminats = c(eliminats, i)}
```

```
## [1] "248 39"
```

Eliminem per tant, la columna 248

```
seMAclean <- seMA[, -eliminats]
```

I fem la imputació dels NA utilitzant el knn que és possiblement el mètode més robust per fer-ho

```
seMAimputed <- seMAclean %>%
```

```
  PomaImpute(method = "knn", zeros_as_na = FALSE, remove_na = FALSE, cutoff = 20)
```

```
## -39 features removed.
```

Després comprovem que no tenim NAs

```
sum(is.na(assay(seMAimputed)))
```

```
## [1] 0
```

Doncs ara, ha podem normalitzar les dades. D'entre els mètodes que es poden utilitzar hem rebutjat els logaritmes perquè amb les dades que tenim ens generen NAs, ek mix_max perquè genera molta variabilitat en les dades en el nostre cas i el box_cox perquè no funciona amb la matriu que tenim.

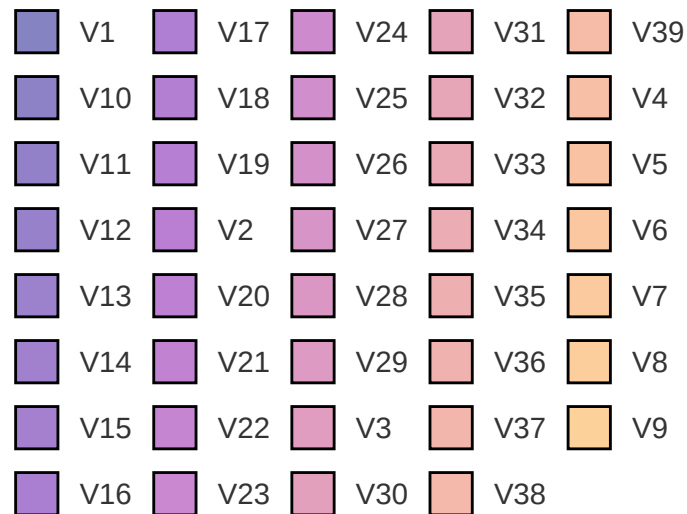
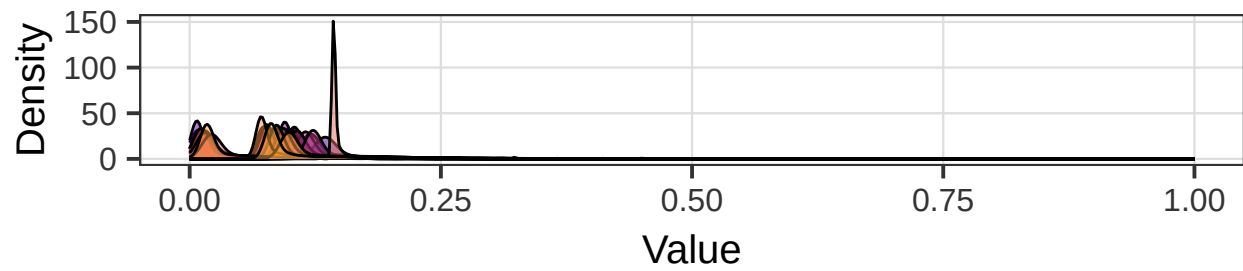
Ens queden els dos mètodes d'escalat i el auto_scaling és el que dona el gràfic més homogeni. Així que ens quedem amb aquest

```
# Ensenyem el resultat de min_max per veure que les funcions de densitat són molt variables
```

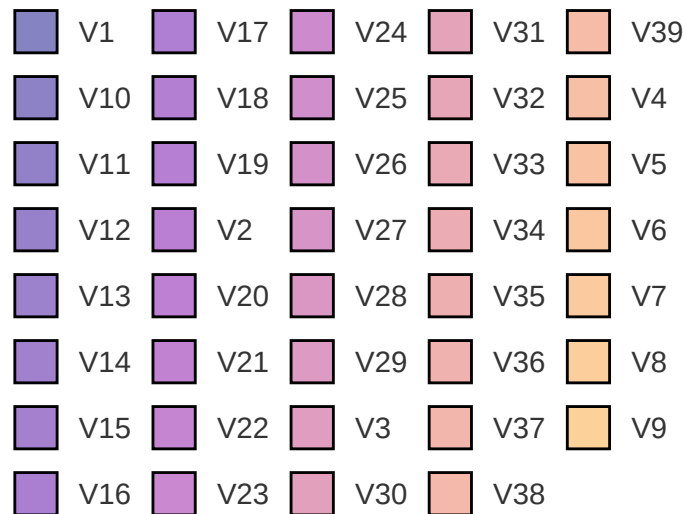
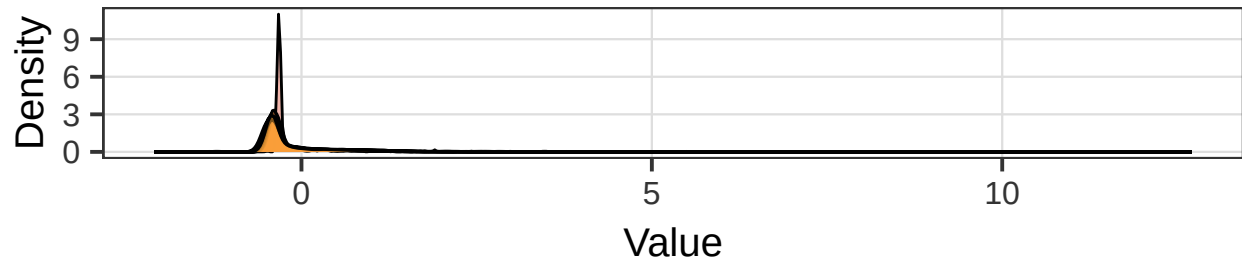
```
seMANormalized <- seMAimputed %>%
```

```
  PomaNorm(method = "min_max")
```

```
PomaDensity(seMANormalized, x = "features")
```



```
# Ensenyem el resultat de min_max per veure que les funcions de densitat són molt variables
seMANormalized <- seMAimputed %>%
  PomaNorm(method = "auto_scaling")
PomaDensity(seMANormalized, x = "features")
```

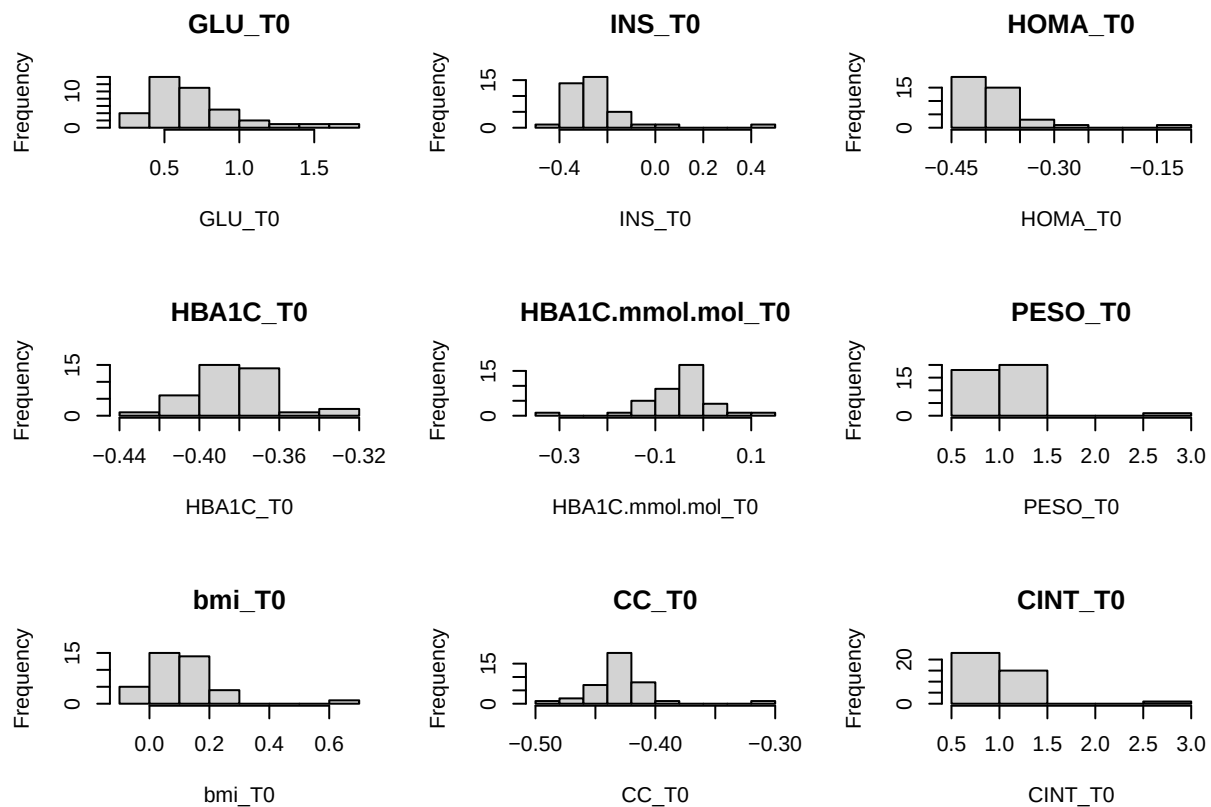


Sembla que hi ha algun outlier, però crec que no és preocupant

5.4. Anàlisi univariant

L'anàlisi univariant no té massa sentit amb una matriu tant gran de dades, però per provar alguna cosa, podem dir que fer alguns histogrames no està de mas. Farem 9 de les variables per veure una mica com es distribueixen després del preprocessament

```
opt <- par(mfrow=c(3,3))
for (i in 1:9)
  hist(assay(seMANormalized)[,i],
       xlab = colData(seMANormalized)[i,2],
       main = colData(seMANormalized)[i,2])
```



```
par(opt)
```

Podem apreciar que les distribucions no són molt homogenies, el que potser implica poques correlacions. Veurem més endavant

6. Reducció de les dimensions

Un pas important ara que hem vist com són les variables pot ser fer un Anàlisi de Components Principals (PCA)). Com que tenim els NAs eliminats i les dades normalitzades, podem fer-lo sense problemes addicionals. Reduim la PCA a 5 components ja que sinó en fa 39 i el resultat s'allarga molt

```
PCAseMA <- prcomp(assay(seMANormalized), scale. = TRUE, rank. = 5)
```

```
PCAseMA
```

```
## Standard deviations (1, ..., p=39):
## [1] 1.893256e+01 7.377935e+00 5.806778e+00 5.366514e+00 5.212884e+00
## [6] 4.421286e+00 4.032098e+00 3.610802e+00 3.464758e+00 2.993985e+00
## [11] 2.905096e+00 2.840228e+00 2.671005e+00 2.633563e+00 2.591891e+00
## [16] 2.550440e+00 2.438817e+00 2.282599e+00 2.222872e+00 2.118927e+00
## [21] 2.067725e+00 2.042961e+00 1.985076e+00 1.889037e+00 1.853185e+00
## [26] 1.802430e+00 1.759200e+00 1.735222e+00 1.711374e+00 1.624513e+00
## [31] 1.531975e+00 1.494274e+00 1.382320e+00 1.374211e+00 1.328333e+00
## [36] 1.224837e+00 1.109147e+00 1.021286e+00 1.161219e-14
##
## Rotation (n x k) = (685 x 5):
##           PC1           PC2           PC3           PC4
## GLU_T0    -1.199624e-02  4.941264e-02 -2.319322e-02 -1.078391e-02
```

## INS_TO	-3.040036e-02	5.672071e-02	-2.721737e-02	2.233584e-02
## HOMA_TO	-4.080096e-02	3.104342e-02	-2.125132e-02	7.222602e-03
## HBA1C_TO	-4.651415e-02	-1.834343e-02	2.008314e-02	-5.565033e-04
## HBA1C.mmol.mol_TO	3.552732e-02	-9.188496e-03	2.104319e-02	4.572940e-04
## PESO_TO	-2.326499e-02	8.125680e-02	-3.630520e-02	-3.611425e-03
## bmi_TO	-2.471270e-02	8.087699e-02	-3.633006e-02	1.303281e-02
## CC_TO	-5.244727e-02	-1.569056e-03	6.832376e-03	6.712868e-03
## CINT_TO	-2.108588e-02	1.005267e-01	-1.260106e-02	2.898376e-02
## CAD_TO	-1.094508e-02	1.031612e-01	-1.546234e-02	-6.292826e-03
## TAD_TO	3.368663e-02	-5.542617e-03	1.880689e-02	-1.076498e-02
## TAS_TO	4.155077e-02	-5.256121e-04	3.309474e-03	-4.469568e-03
## TG_TO	4.626877e-03	2.775474e-02	-5.204236e-02	5.830628e-02
## COL_TO	-3.209452e-03	4.890035e-02	-1.392578e-02	2.888548e-02
## LDL_TO	-9.495424e-03	6.748697e-02	-1.252003e-02	2.204182e-02
## HDL_TO	6.530946e-04	-3.647308e-02	2.226752e-02	-1.236797e-02
## VLDL_TO	-4.486175e-03	2.630072e-02	-5.128724e-02	5.894473e-02
## PCR_TO	-1.594345e-02	-2.534249e-02	2.709684e-02	-1.312719e-02
## LEP_TO	-2.977204e-02	5.974349e-02	-1.459339e-02	5.927173e-03
## ADIPO_TO	-2.157399e-02	5.445112e-03	-1.229830e-02	-2.448064e-02
## GOT_TO	-1.097243e-02	3.930316e-03	3.068829e-02	3.609539e-03
## GPT_TO	-1.042910e-02	6.848081e-03	-1.324231e-02	-8.294370e-03
## GGT_TO	-2.106519e-02	5.718606e-02	-4.429951e-02	4.709703e-02
## URICO_TO	-4.690040e-02	3.257911e-02	-1.050633e-02	2.145387e-02
## CREAT_TO	-5.238177e-02	-4.462782e-03	7.125217e-03	3.583363e-04
## UREA_TO	-1.102555e-02	4.186594e-02	-8.738753e-03	3.110859e-02
## HIERRO_TO	-3.876584e-03	4.665753e-02	-5.411954e-03	2.833490e-02
## TRANSF_TO	3.990148e-02	-6.735974e-03	1.411681e-02	2.756888e-02
## FERR_TO	1.100355e-02	2.202379e-02	1.604286e-02	1.859058e-03
## Ile_TO	-9.979157e-03	4.347839e-02	-8.025407e-02	3.102797e-02
## Leu_TO	-1.901120e-03	3.243955e-02	-8.925402e-02	2.524131e-02
## Val_TO	1.370998e-03	2.254413e-02	-9.141408e-02	3.520201e-02
## Ala_TO	-5.657694e-03	5.095690e-02	-1.196257e-01	2.891007e-02
## Pro_TO	-2.375758e-02	4.844247e-02	-8.208939e-02	-1.149389e-02
## Gly_TO	-1.004746e-02	3.877661e-02	-6.471853e-02	-2.918055e-02
## Ser_TO	-1.665486e-02	5.123520e-02	-7.174336e-02	-1.949394e-02
## Trp_TO	-1.706001e-02	1.060458e-02	-9.699474e-02	2.718742e-02
## Phe_TO	-8.719772e-03	4.965718e-02	-1.062346e-01	1.202598e-04
## Met_TO	2.364641e-02	-8.029138e-02	4.323907e-02	-2.661267e-02
## Orn_TO	-9.794162e-03	4.300912e-03	-8.253731e-02	2.358071e-02
## Arg_TO	-8.207693e-03	7.242484e-02	-9.691286e-02	5.283762e-02
## His_TO	-1.961687e-02	3.543715e-02	-1.080403e-01	-6.512983e-03
## Asn_TO	-2.278992e-02	-1.581569e-02	-2.588730e-02	-2.529066e-02
## Asp_TO	-2.454699e-02	7.611141e-02	-6.003059e-02	3.112978e-02
## Glu_TO	-1.891976e-02	9.132817e-02	-6.796917e-02	5.490545e-02
## Gln_TO	-6.384554e-03	-5.760449e-02	-2.247765e-02	-7.017922e-02
## Cit_TO	-3.067717e-02	4.560629e-02	-3.427953e-02	3.033582e-02
## Tyr_TO	-1.926940e-02	2.538624e-02	-1.016609e-01	1.033410e-02
## Thr_TO	-3.264978e-03	-2.056393e-02	-4.111030e-02	1.093298e-02
## Lys_TO	-6.434112e-03	1.624795e-02	-1.015303e-01	4.084040e-02
## Creatinine_TO	-1.030747e-02	5.069871e-02	-3.989496e-02	-1.454879e-03
## Kynurenine_TO	-4.960725e-02	-3.669416e-03	-2.642139e-02	-9.883789e-04
## Putrescine_TO	-7.308058e-03	-6.104151e-04	-1.309582e-02	5.102346e-03
## Sarcosine_TO	-3.160741e-02	4.123728e-02	-1.020420e-02	2.945476e-02
## Serotonin_TO	-7.413211e-03	-7.621286e-03	4.666980e-02	5.402820e-03

## Taurine_TO	-1.772370e-02	3.613755e-02	-3.522009e-02	1.739916e-02
## SDMA_TO	-5.241650e-02	-2.826580e-03	3.299265e-03	6.957672e-04
## C0_TO	-2.576613e-02	4.587907e-02	-8.069356e-02	4.560331e-02
## C2_TO	-2.218579e-02	-2.907379e-02	3.813234e-02	-2.346379e-02
## C3.OH_TO	-5.247407e-02	-8.697286e-03	6.063526e-03	1.223400e-03
## C6..C4.1.DC._TO	3.267974e-03	3.041575e-02	4.521626e-02	1.201054e-02
## C5.DC..C6.OH._TO	-3.565798e-02	-1.456302e-02	3.969868e-03	1.830535e-02
## C7.DC_TO	-3.495627e-02	-5.186282e-03	6.317214e-03	-5.121927e-03
## C8_TO	-5.250656e-02	-7.142245e-03	5.207529e-03	2.284466e-03
## C10_TO	-5.251524e-02	-5.433485e-03	3.792803e-03	2.205449e-03
## C10.1_TO	-5.249195e-02	-8.578609e-03	4.615177e-03	1.155361e-03
## C10.2_TO	-5.245744e-02	-8.645811e-03	6.548846e-03	1.100308e-03
## C14.1_TO	-5.249986e-02	-8.257121e-03	5.865208e-03	1.783103e-03
## C14.2_TO	-5.247850e-02	-8.609041e-03	6.262901e-03	1.412700e-03
## C16.1_TO	-5.248726e-02	-8.523857e-03	6.458853e-03	1.373352e-03
## C16.2_TO	-5.246910e-02	-9.095494e-03	6.594646e-03	1.459950e-03
## C16.2.OH_TO	-5.247213e-02	-9.103851e-03	6.569942e-03	1.380622e-03
## C18.1_TO	-5.253721e-02	-7.752416e-03	5.969628e-03	1.142369e-03
## C18.1.OH_TO	-5.246717e-02	-9.145540e-03	6.585084e-03	1.396145e-03
## C18.2_TO	-5.248329e-02	-8.754111e-03	6.588226e-03	1.652488e-03
## lysoPC.a.C16.0_TO	-1.888892e-02	7.119173e-02	-1.024627e-01	3.386555e-02
## lysoPC.a.C16.1_TO	-4.645715e-02	2.996157e-02	-5.568621e-02	1.882351e-02
## lysoPC.a.C17.0_TO	-5.085825e-02	1.727304e-02	-1.904967e-02	1.555739e-02
## lysoPC.a.C18.0_TO	-1.950736e-02	5.616350e-02	-1.097307e-01	3.486232e-02
## lysoPC.a.C18.1_TO	-2.845384e-02	5.606669e-02	-7.789242e-02	2.511448e-02
## lysoPC.a.C18.2_TO	-2.672044e-02	1.336755e-02	-8.544482e-02	9.233331e-03
## lysoPC.a.C20.3_TO	-4.491626e-02	2.014642e-03	-5.759618e-02	2.154599e-02
## lysoPC.a.C20.4_TO	-3.186501e-02	3.680045e-02	-4.321636e-02	1.251284e-02
## lysoPC.a.C24.0_TO	-5.259933e-02	-4.423072e-03	1.436010e-03	2.071438e-03
## lysoPC.a.C26.0_TO	-4.635546e-02	-8.375227e-04	-1.651850e-02	2.296390e-02
## lysoPC.a.C26.1_TO	-5.260177e-02	-4.112828e-03	3.559603e-03	7.598541e-04
## lysoPC.a.C28.0_TO	-3.820598e-02	-1.176691e-02	-3.463173e-02	1.541886e-02
## lysoPC.a.C28.1_TO	-5.268437e-02	9.811227e-04	-1.621543e-03	3.456510e-03
## PC.aa.C24.0_TO	-3.450394e-02	1.405759e-02	-1.144507e-02	-1.788179e-02
## PC.aa.C28.1_TO	-5.090210e-02	-5.572934e-03	-8.989270e-03	1.676002e-03
## PC.aa.C30.0_TO	-5.076110e-02	-1.062761e-02	-4.171637e-03	5.860590e-03
## PC.aa.C32.0_TO	-4.047902e-02	-1.049841e-02	-2.208971e-02	-5.985915e-03
## PC.aa.C32.1_TO	-2.933937e-02	-4.226707e-03	-5.208939e-02	1.485087e-02
## PC.aa.C32.3_TO	-5.254305e-02	-8.265740e-03	4.482921e-03	2.645503e-03
## PC.aa.C34.1_TO	-4.679961e-03	-2.399894e-03	-5.577084e-02	2.887736e-02
## PC.aa.C34.2_TO	-5.839240e-03	-5.785504e-02	-3.948971e-02	3.145463e-02
## PC.aa.C34.3_TO	-3.396562e-02	-2.782546e-02	-5.344318e-02	2.916803e-02
## PC.aa.C34.4_TO	-5.193914e-02	-1.288689e-02	-9.518723e-04	7.521392e-03
## PC.aa.C36.0_TO	-5.113143e-02	1.940279e-02	-1.720350e-02	1.913615e-02
## PC.aa.C36.1_TO	-2.407247e-02	-2.667213e-03	-7.121721e-02	5.948390e-02
## PC.aa.C36.2_TO	-8.060236e-03	-4.625739e-02	-6.466646e-02	3.938564e-02
## PC.aa.C36.3_TO	-6.934274e-03	-4.507168e-02	-5.754152e-02	4.404455e-02
## PC.aa.C36.4_TO	7.269459e-03	-1.944508e-02	-3.726738e-03	2.456261e-02
## PC.aa.C36.5_TO	-2.377410e-02	1.315630e-02	-1.724130e-02	5.942156e-02
## PC.aa.C38.0_TO	-4.909721e-02	1.107830e-02	-2.123144e-02	1.879520e-02
## PC.aa.C38.1_TO	-4.226607e-02	3.070666e-02	-3.542939e-04	3.360842e-02
## PC.aa.C38.3_TO	-7.449456e-03	-2.944195e-02	-8.810957e-02	4.505764e-02
## PC.aa.C38.4_TO	4.819443e-03	-1.588009e-02	-3.221724e-02	1.804556e-02
## PC.aa.C38.5_TO	-8.793357e-03	-1.005540e-02	-2.803464e-02	3.987593e-02

## PC.aa.C38.6_TO	3.682175e-03	-5.666075e-03	-2.843888e-03	1.717111e-02
## PC.aa.C40.1_TO	-3.532944e-02	-7.802958e-03	-5.990080e-02	3.450067e-02
## PC.aa.C40.2_TO	-4.948016e-02	2.974843e-02	-2.947529e-02	2.070395e-02
## PC.aa.C40.3_TO	-5.078606e-02	2.243592e-02	-2.445202e-02	1.596725e-02
## PC.aa.C40.4_TO	-4.967989e-02	1.463772e-02	-4.116538e-02	2.311102e-02
## PC.aa.C40.5_TO	-4.508810e-02	-1.838723e-02	-3.328628e-02	1.524805e-02
## PC.aa.C40.6_TO	-6.193391e-03	-4.759122e-03	-2.173514e-02	3.784442e-02
## PC.aa.C42.0_TO	-5.263114e-02	3.232547e-03	-4.267086e-03	5.772590e-03
## PC.aa.C42.1_TO	-5.270876e-02	1.840187e-03	-2.633491e-03	5.705166e-03
## PC.aa.C42.2_TO	-5.235307e-02	9.994660e-03	-1.034820e-02	9.934093e-03
## PC.aa.C42.4_TO	-5.233352e-02	7.367583e-03	-1.202856e-02	9.351755e-03
## PC.aa.C42.5_TO	-5.269167e-02	-1.287353e-03	-3.441852e-03	4.735824e-03
## PC.aa.C42.6_TO	-4.304304e-02	-4.360666e-03	-1.269944e-02	1.956059e-02
## PC.ae.C30.0_TO	-5.254257e-02	-7.763565e-03	3.942071e-03	1.377721e-03
## PC.ae.C32.1_TO	-5.134015e-02	-1.054594e-02	-4.191666e-03	-3.771758e-03
## PC.ae.C32.2_TO	-5.254671e-02	-7.489553e-03	4.855983e-04	2.477391e-03
## PC.ae.C34.0_TO	-5.252130e-02	-7.244882e-03	4.246375e-03	3.879746e-03
## PC.ae.C34.1_TO	-4.894199e-02	-1.066988e-02	-7.958685e-03	5.977845e-03
## PC.ae.C34.2_TO	-3.815767e-02	-3.850930e-02	-2.519837e-02	1.399627e-02
## PC.ae.C34.3_TO	-4.339116e-02	-3.679685e-02	-1.527740e-02	6.648265e-03
## PC.ae.C36.0_TO	-5.256366e-02	2.609808e-03	-5.873801e-03	8.648078e-03
## PC.ae.C36.1_TO	-3.954970e-02	6.158932e-02	-5.052192e-02	3.778079e-02
## PC.ae.C36.2_TO	-4.722815e-02	1.147999e-02	-1.985515e-02	2.980044e-02
## PC.ae.C36.3_TO	-4.564989e-02	-2.815843e-02	-1.852507e-02	1.615099e-02
## PC.ae.C36.4_TO	-3.142224e-02	-3.911879e-02	-2.895686e-02	1.854032e-02
## PC.ae.C36.5_TO	-3.810759e-02	-3.191308e-02	-1.404708e-02	1.430391e-02
## PC.ae.C38.0_TO	-5.176857e-02	-5.287911e-03	-5.684053e-03	1.273141e-02
## PC.ae.C38.2_TO	-3.518049e-02	5.861955e-02	-6.408075e-02	3.593753e-02
## PC.ae.C38.3_TO	-4.089493e-02	5.243691e-02	-5.363895e-02	5.026145e-02
## PC.ae.C38.4_TO	-4.569485e-02	2.625039e-03	-2.448249e-02	2.056411e-02
## PC.ae.C38.5_TO	-3.192526e-02	-2.723012e-02	-1.504464e-02	2.332015e-03
## PC.ae.C38.6_TO	-4.613645e-02	-2.173110e-02	-5.003642e-03	1.414507e-02
## PC.ae.C40.1_TO	-5.084815e-02	2.284226e-02	-2.106949e-02	1.964880e-02
## PC.ae.C40.2_TO	-4.710051e-02	3.631939e-02	-3.784293e-02	2.692484e-02
## PC.ae.C40.3_TO	-4.020022e-02	5.202573e-02	-5.975111e-02	4.330442e-02
## PC.ae.C40.4_TO	-4.713499e-02	3.459666e-02	-4.848859e-02	2.895913e-02
## PC.ae.C40.5_TO	-4.608866e-02	4.094891e-02	-4.734619e-02	3.106280e-02
## PC.ae.C40.6_TO	-5.133905e-02	-7.497071e-03	4.333479e-05	7.503292e-03
## PC.ae.C42.1_TO	-5.172954e-02	1.700410e-02	-1.623628e-02	1.361134e-02
## PC.ae.C42.2_TO	-5.214331e-02	1.293312e-02	-1.249491e-02	1.188298e-02
## PC.ae.C42.3_TO	-5.140818e-02	1.829603e-02	-1.933481e-02	1.482275e-02
## PC.ae.C42.4_TO	-5.233265e-02	5.381443e-03	-1.570495e-02	9.153817e-03
## PC.ae.C42.5_TO	-5.075374e-02	1.856278e-02	-2.889258e-02	1.420372e-02
## PC.ae.C44.3_TO	-5.262260e-02	2.564179e-03	-4.501264e-03	6.423447e-03
## PC.ae.C44.4_TO	-5.264653e-02	-3.518964e-03	-8.844004e-04	3.910610e-03
## PC.ae.C44.5_TO	-5.078459e-02	-1.029710e-02	2.150019e-03	-1.480206e-03
## PC.ae.C44.6_TO	-5.167694e-02	-1.145469e-02	-1.209864e-03	-4.265336e-04
## SM.OH.C14.1_TO	-4.928159e-02	-1.530486e-02	9.313234e-03	-5.387356e-03
## SM.OH.C16.1_TO	-5.181957e-02	-8.689912e-03	1.020446e-02	1.710025e-03
## SM.OH.C22.1_TO	-3.615453e-02	-4.026414e-02	6.899994e-03	1.878689e-02
## SM.OH.C22.2_TO	-4.666995e-02	3.342502e-03	-1.296481e-02	1.391046e-02
## SM.OH.C24.1_TO	-5.226009e-02	-1.048512e-02	8.167850e-03	2.521347e-03
## SM.C16.0_TO	-7.828120e-03	-8.261959e-03	-3.558635e-03	-1.574621e-02
## SM.C16.1_TO	-3.170526e-02	-2.404167e-02	-1.615961e-02	-1.583856e-02

## SM.C18.0_T0	-2.901498e-02	1.082833e-02	1.581796e-02	2.361674e-03
## SM.C18.1_T0	-3.905315e-02	-3.637680e-03	8.261195e-03	-6.774493e-03
## SM.C20.2_T0	-5.245883e-02	-9.913957e-03	6.340898e-03	1.655250e-03
## SM.C24.0_T0	-3.504443e-02	-1.304466e-02	-3.389091e-02	1.930640e-02
## SM.C24.1_T0	-1.898667e-02	1.945600e-02	5.773657e-03	3.296389e-03
## MEDDM_T2	-5.245502e-02	-9.564885e-03	6.632371e-03	1.080576e-03
## MEDCOL_T2	-5.245502e-02	-9.564885e-03	6.632371e-03	1.080576e-03
## MEDINF_T2	-5.211366e-02	-1.071897e-02	8.022690e-03	4.077391e-03
## MEDHTA_T2	-5.191703e-02	-6.650130e-03	4.399277e-04	8.783626e-03
## GLU_T2	3.825544e-02	4.252139e-03	8.247871e-03	6.010433e-03
## INS_T2	-4.434399e-04	-1.904866e-02	1.892014e-02	-5.602655e-02
## HOMA_T2	-3.988365e-02	-1.687539e-02	1.363804e-02	-2.900605e-02
## HBA1C_T2	-4.792830e-02	-1.739073e-02	2.556816e-02	-3.765267e-03
## HBA1C.mmol.mol_T2	3.171862e-02	-1.119653e-02	5.174481e-02	-1.201423e-02
## PESO_T2	3.380214e-02	-1.080327e-02	-2.174437e-02	-3.614014e-02
## bmi_T2	3.172695e-02	-1.218128e-02	-2.219040e-02	-1.703664e-02
## CC_T2	-5.216416e-02	-9.795423e-03	8.209750e-03	2.364884e-03
## CINT_T2	4.225017e-02	-1.881668e-03	-4.082745e-03	-4.780645e-03
## CAD_T2	4.179569e-02	-1.463169e-02	-1.059582e-02	-2.620972e-02
## TAD_T2	4.150739e-02	7.100272e-03	-2.455751e-02	2.022245e-02
## TAS_T2	4.103215e-02	8.194423e-03	-1.866632e-02	1.211508e-03
## TG_T2	2.816995e-02	-5.140429e-03	2.624929e-02	8.348114e-03
## COL_T2	3.435067e-02	-1.191990e-02	5.950666e-02	3.593478e-03
## LDL_T2	3.493637e-02	6.368421e-04	5.580306e-02	1.864162e-02
## HDL_T2	1.345175e-02	-2.688160e-02	6.069721e-02	-3.438750e-03
## VLDL_T2	9.435087e-03	-2.811919e-02	3.547613e-02	-1.640362e-02
## PCR_T2	-1.412898e-02	-7.153529e-04	4.185992e-02	-2.697978e-02
## LEP_T2	1.146354e-02	-1.097647e-02	1.987033e-02	2.800129e-03
## ADIPO_T2	-9.355925e-03	1.016337e-03	-1.348928e-02	-1.676050e-02
## GOT_T2	2.360281e-03	-4.586665e-02	1.356174e-02	1.720447e-02
## GPT_T2	7.162270e-03	-2.884920e-02	-5.487717e-03	1.160603e-02
## GGT_T2	7.795501e-03	-2.681696e-02	2.554680e-02	1.852152e-02
## URICO_T2	-3.060967e-02	-2.809284e-02	2.956216e-03	-3.039250e-02
## CREAT_T2	-5.200608e-02	-1.218551e-02	7.035255e-03	-3.953076e-03
## UREA_T2	1.408107e-02	1.329286e-02	7.416350e-02	-2.894384e-02
## HIERRO_T2	1.517783e-02	-1.184607e-02	1.936847e-02	6.521345e-02
## TRANSF_T2	4.199475e-02	-1.246983e-02	2.440201e-02	2.761878e-03
## FERR_T2	7.675456e-03	1.469931e-02	-8.725266e-03	-1.736711e-02
## Ile_T2	-8.621274e-03	5.497743e-02	-2.719281e-03	-1.001336e-01
## Leu_T2	-7.991628e-03	5.891425e-02	4.834765e-04	-1.001346e-01
## Val_T2	-1.093656e-02	6.777186e-02	1.628445e-02	-1.015320e-01
## Ala_T2	-1.674712e-02	7.820676e-02	3.684546e-02	-5.876309e-02
## Pro_T2	-1.612979e-02	6.755335e-02	1.536146e-02	-8.844213e-02
## Gly_T2	-5.530842e-03	3.553334e-02	4.204593e-02	-7.628765e-02
## Ser_T2	-3.100467e-03	4.105752e-02	2.107735e-02	-8.835778e-02
## Trp_T2	-2.766754e-03	6.757765e-02	7.508522e-02	-8.954113e-02
## Phe_T2	-5.692661e-03	6.578627e-02	2.704765e-02	-9.437132e-02
## Met_T2	-1.747946e-02	3.089153e-03	4.662582e-02	-1.477892e-02
## Orn_T2	-3.148348e-03	4.774206e-02	3.655536e-02	-7.391611e-02
## Arg_T2	-8.224978e-03	9.950650e-02	5.251903e-02	-3.832805e-02
## His_T2	-1.376341e-02	9.003043e-02	2.209472e-02	-8.717891e-02
## Asn_T2	-2.652617e-02	6.025220e-02	6.064235e-02	-7.983231e-02
## Asp_T2	-9.997523e-04	3.315654e-02	3.544431e-02	-6.710315e-03
## Glu_T2	7.175733e-03	6.425496e-02	-7.479546e-02	3.022789e-02

## Gln_T2	-1.947893e-02	3.329075e-02	8.014939e-02	-7.136711e-02
## Cit_T2	-2.384681e-02	6.593545e-02	5.819803e-02	-5.666937e-02
## Tyr_T2	-1.792655e-02	7.871440e-02	4.156426e-02	-6.984888e-02
## Thr_T2	-1.433721e-02	3.885229e-02	2.722568e-02	-8.101006e-02
## Lys_T2	-6.023937e-03	7.755619e-02	4.519433e-02	-8.510239e-02
## Creatinine_T2	-7.893733e-03	6.571092e-02	3.904550e-02	-8.615072e-02
## Kynurenine_T2	-5.067728e-02	4.045360e-03	2.062023e-02	-2.315684e-02
## Putrescine_T2	3.431822e-03	2.780524e-02	-1.475363e-02	9.083907e-03
## Sarcosine_T2	-2.618470e-02	3.965990e-02	4.400402e-02	-3.359136e-02
## Serotonin_T2	-1.023620e-02	9.753314e-03	-1.196136e-02	-3.510861e-02
## Taurine_T2	-1.174743e-02	3.227001e-02	9.268447e-03	-3.186203e-03
## SDMA_T2	-5.156382e-02	1.879322e-03	1.290757e-02	-1.335975e-02
## C0_T2	-1.308114e-02	7.750648e-02	4.646985e-02	-1.763247e-02
## C2_T2	-3.077317e-02	-1.270500e-02	1.518447e-04	-2.498897e-02
## C3.OH_T2	-5.246289e-02	-8.721361e-03	7.433831e-03	-4.104634e-04
## C6..C4.1.DC._T2	-8.794011e-05	1.910149e-02	-3.701148e-02	7.061459e-02
## C5.DC..C6.OH._T2	-3.966257e-02	-7.223369e-05	1.488706e-02	2.432386e-02
## C7.DC_T2	-3.945212e-02	-1.980068e-02	2.891609e-02	-2.036034e-02
## C8_T2	-5.247319e-02	-5.835348e-03	9.043459e-03	-1.731781e-03
## C10_T2	-5.246485e-02	-5.304002e-03	8.694260e-03	-3.987744e-03
## C10.1_T2	-5.247819e-02	-8.349250e-03	6.843062e-03	-1.389674e-03
## C10.2_T2	-5.247311e-02	-8.721554e-03	8.374141e-03	8.508383e-05
## C14.1_T2	-5.246604e-02	-8.142849e-03	7.353581e-03	1.867183e-04
## C14.2_T2	-5.244983e-02	-8.934934e-03	7.540346e-03	5.834426e-04
## C16.1_T2	-5.246639e-02	-8.892285e-03	7.453149e-03	9.577030e-04
## C16.2_T2	-5.246115e-02	-9.330009e-03	6.968043e-03	1.079576e-03
## C16.2.OH_T2	-5.246307e-02	-9.292363e-03	6.853309e-03	1.052735e-03
## C18.1_T2	-5.247788e-02	-8.059194e-03	7.173419e-03	-1.154825e-03
## C18.1.OH_T2	-5.245963e-02	-9.360578e-03	7.095768e-03	9.061943e-04
## C18.2_T2	-5.244010e-02	-9.203304e-03	7.137599e-03	3.176886e-04
## lysoPC.a.C14.0_T2	-2.281826e-02	-2.099876e-02	4.869040e-02	-6.003401e-02
## lysoPC.a.C16.0_T2	3.493555e-03	7.448962e-02	1.829048e-02	-6.285216e-02
## lysoPC.a.C16.1_T2	-4.670242e-02	2.390311e-02	2.526679e-02	-3.617971e-02
## lysoPC.a.C17.0_T2	-5.072213e-02	1.749478e-02	1.964668e-02	-1.388903e-02
## lysoPC.a.C18.0_T2	-5.873949e-03	9.003806e-02	4.289960e-02	-3.330789e-02
## lysoPC.a.C18.1_T2	-7.168029e-03	6.562473e-02	5.350483e-02	-8.218161e-02
## lysoPC.a.C18.2_T2	-1.281283e-02	5.720073e-02	6.436753e-02	-6.609383e-02
## lysoPC.a.C20.3_T2	-4.856328e-02	2.413333e-02	2.823329e-02	-1.929142e-02
## lysoPC.a.C20.4_T2	-3.083156e-02	3.708094e-02	2.702869e-02	-6.382399e-02
## lysoPC.a.C24.0_T2	-5.250876e-02	-5.785631e-03	7.645641e-03	8.490239e-04
## lysoPC.a.C26.0_T2	-3.694677e-02	-1.823682e-03	8.127999e-03	-2.731974e-02
## lysoPC.a.C26.1_T2	-5.242601e-02	-8.338033e-03	6.106086e-03	-1.056458e-03
## lysoPC.a.C28.0_T2	-2.936705e-02	1.280549e-02	1.984245e-02	-3.555226e-02
## lysoPC.a.C28.1_T2	-5.241746e-02	-4.941967e-03	6.137211e-03	-8.752074e-04
## PC.aa.C24.0_T2	-1.427309e-02	-3.239714e-02	2.729680e-02	-2.494807e-02
## PC.aa.C28.1_T2	-5.132681e-02	9.440831e-03	3.099406e-02	-1.031314e-02
## PC.aa.C30.0_T2	-5.202883e-02	-1.455034e-03	1.700220e-02	-8.123360e-03
## PC.aa.C32.0_T2	-3.869823e-02	2.687094e-02	5.608107e-02	-5.442115e-02
## PC.aa.C32.1_T2	-3.876949e-02	3.663324e-02	3.865157e-02	-3.515520e-02
## PC.aa.C32.3_T2	-5.253778e-02	-7.642949e-03	6.702165e-03	-4.443329e-04
## PC.aa.C34.1_T2	-6.871676e-03	6.605999e-02	9.276728e-02	-7.690553e-02
## PC.aa.C34.2_T2	-2.109368e-02	6.180995e-02	9.372387e-02	-4.502574e-02
## PC.aa.C34.3_T2	-3.948838e-02	4.385576e-02	5.840761e-02	-2.567210e-02
## PC.aa.C34.4_T2	-5.250617e-02	-4.972344e-03	1.066010e-02	-1.586721e-03

## PC.aa.C36.0_T2	-5.175205e-02	6.067581e-03	7.172530e-03	-6.574652e-03
## PC.aa.C36.1_T2	-2.360321e-02	8.036915e-02	7.265451e-02	-3.386197e-02
## PC.aa.C36.2_T2	-1.459649e-02	7.477894e-02	9.525628e-02	-1.698838e-02
## PC.aa.C36.3_T2	-1.460511e-02	7.605173e-02	8.745500e-02	-3.869726e-02
## PC.aa.C36.4_T2	-2.018011e-02	3.026035e-02	5.029445e-02	-5.499788e-02
## PC.aa.C36.5_T2	-2.966238e-02	3.767907e-02	4.615501e-02	-4.347601e-02
## PC.aa.C38.0_T2	-5.023731e-02	2.264437e-02	2.931214e-02	-1.909040e-02
## PC.aa.C38.1_T2	-4.530134e-02	2.682410e-02	-2.860314e-04	-2.944812e-02
## PC.aa.C38.3_T2	-2.349020e-02	8.168681e-02	7.646186e-02	-2.231953e-02
## PC.aa.C38.4_T2	-1.962151e-02	6.048277e-02	8.261177e-02	-5.319342e-02
## PC.aa.C38.5_T2	-2.422760e-02	5.609569e-02	8.694400e-02	-6.374079e-02
## PC.aa.C38.6_T2	-1.405830e-02	4.953124e-02	6.309218e-02	-3.780808e-02
## PC.aa.C40.1_T2	-2.775708e-02	3.940623e-02	7.435345e-03	-5.730204e-02
## PC.aa.C40.2_T2	-5.042860e-02	6.634111e-03	-1.657985e-02	7.106866e-03
## PC.aa.C40.3_T2	-5.124411e-02	4.023766e-03	-5.621997e-03	-1.139779e-03
## PC.aa.C40.4_T2	-5.049400e-02	2.448560e-02	1.043348e-02	-7.141418e-03
## PC.aa.C40.5_T2	-4.539757e-02	3.198962e-02	5.355546e-02	-2.436583e-02
## PC.aa.C40.6_T2	-2.352882e-02	6.759138e-02	6.843537e-02	-2.036261e-02
## PC.aa.C42.0_T2	-5.253554e-02	-3.916966e-03	5.710044e-03	-2.719226e-03
## PC.aa.C42.1_T2	-5.252381e-02	-5.140283e-03	5.328132e-03	-1.722765e-03
## PC.aa.C42.2_T2	-5.233557e-02	-2.056739e-03	-3.032401e-03	9.581801e-04
## PC.aa.C42.4_T2	-5.248732e-02	-2.398640e-03	-3.353148e-03	4.047450e-03
## PC.aa.C42.5_T2	-5.260377e-02	-4.255415e-03	4.630732e-03	2.678030e-04
## PC.aa.C42.6_T2	-4.149299e-02	-7.351144e-03	2.247671e-02	-1.793776e-02
## PC.ae.C30.0_T2	-3.866764e-02	3.953976e-03	1.315170e-02	-3.710400e-02
## PC.ae.C32.1_T2	-5.158453e-02	6.287398e-03	2.350916e-02	-1.888153e-02
## PC.ae.C32.2_T2	-5.256756e-02	-3.948619e-03	1.068639e-02	-4.495941e-03
## PC.ae.C34.0_T2	-5.254033e-02	-3.562469e-03	1.103251e-02	-1.613118e-03
## PC.ae.C34.1_T2	-4.645693e-02	3.031314e-02	4.895663e-02	-3.711287e-02
## PC.ae.C34.2_T2	-4.617194e-02	3.563224e-02	5.082014e-02	-3.033178e-02
## PC.ae.C34.3_T2	-4.844283e-02	2.431023e-02	4.018361e-02	-1.851495e-02
## PC.ae.C36.0_T2	-5.258760e-02	3.198468e-03	7.374210e-03	-5.493374e-03
## PC.ae.C36.1_T2	-3.117258e-02	4.693675e-02	-3.237397e-02	2.902262e-03
## PC.ae.C36.2_T2	-4.512749e-02	3.938198e-02	4.617977e-02	-2.463510e-02
## PC.ae.C36.3_T2	-5.032515e-02	1.799711e-02	3.413767e-02	-1.825086e-02
## PC.ae.C36.4_T2	-4.494408e-02	3.281381e-02	5.986622e-02	-4.009632e-02
## PC.ae.C36.5_T2	-4.560357e-02	3.248087e-02	5.238353e-02	-3.456805e-02
## PC.ae.C38.0_T2	-5.193794e-02	8.088385e-03	1.926391e-02	-7.537202e-03
## PC.ae.C38.2_T2	-4.133954e-02	3.041475e-02	-3.077215e-02	1.035068e-02
## PC.ae.C38.3_T2	-3.854888e-02	3.658070e-02	-4.625893e-02	1.518008e-02
## PC.ae.C38.4_T2	-4.751978e-02	3.432325e-02	4.018823e-02	-3.478686e-02
## PC.ae.C38.5_T2	-4.018721e-02	3.833141e-02	6.642050e-02	-5.865205e-02
## PC.ae.C38.6_T2	-4.949739e-02	2.061775e-02	4.156293e-02	-2.144402e-02
## PC.ae.C40.1_T2	-5.075761e-02	1.149852e-02	-2.444890e-03	-3.679518e-03
## PC.ae.C40.2_T2	-5.148151e-02	1.317231e-02	1.771201e-03	-1.264436e-03
## PC.ae.C40.3_T2	-4.283186e-02	2.643150e-02	-4.399597e-02	1.635151e-02
## PC.ae.C40.4_T2	-4.962281e-02	2.384091e-02	-1.290434e-02	-5.914266e-03
## PC.ae.C40.5_T2	-4.731579e-02	3.100539e-02	-3.163526e-02	1.496497e-03
## PC.ae.C40.6_T2	-5.135826e-02	1.550029e-02	2.686007e-02	-1.532806e-02
## PC.ae.C42.1_T2	-5.189516e-02	1.142358e-03	-4.632406e-03	2.442960e-03
## PC.ae.C42.2_T2	-5.205024e-02	3.838136e-04	6.813092e-04	-2.497777e-03
## PC.ae.C42.3_T2	-5.176138e-02	6.231921e-03	-6.203187e-03	5.002887e-04
## PC.ae.C42.4_T2	-5.244812e-02	1.292784e-03	6.611616e-03	-3.296204e-03
## PC.ae.C42.5_T2	-5.161280e-02	1.297242e-02	2.214389e-03	-1.455270e-02

## PC.ae.C44.3_T2	-4.360104e-02	4.895102e-03	2.906923e-02	1.629672e-02
## PC.ae.C44.4_T2	-5.251256e-02	-5.460471e-03	6.536077e-03	-1.266908e-03
## PC.ae.C44.5_T2	-5.106780e-02	6.549883e-04	2.490225e-02	-2.180662e-02
## PC.ae.C44.6_T2	-5.192468e-02	-4.793165e-03	1.721883e-02	-1.264488e-02
## SM..OH..C14.1_T2	-4.880184e-02	1.826323e-02	4.003102e-02	-1.380669e-02
## SM..OH..C16.1_T2	-5.176974e-02	5.650898e-03	2.461918e-02	-1.938129e-03
## SM..OH..C22.1_T2	-4.818457e-02	3.029608e-02	4.304018e-02	-3.241965e-03
## SM..OH..C22.2_T2	-4.752607e-02	3.605125e-02	3.492114e-02	-9.298929e-03
## SM..OH..C24.1_T2	-5.255810e-02	-3.864650e-03	1.121992e-02	1.417137e-04
## SM.C16.0_T2	-1.835175e-02	6.548734e-02	9.266550e-02	-2.850688e-02
## SM.C16.1_T2	-3.935709e-02	3.352410e-02	4.735321e-02	-2.714570e-02
## SM.C18.0_T2	-2.653149e-02	3.169499e-02	6.981412e-02	3.677801e-03
## SM.C18.1_T2	-3.952526e-02	1.579564e-02	4.474842e-02	-9.096202e-03
## SM.C20.2_T2	-5.248856e-02	-8.014355e-03	8.616849e-03	2.762188e-05
## SM.C24.0_T2	-4.357222e-02	4.788717e-02	4.456885e-02	-4.712070e-03
## SM.C24.1_T2	-2.191563e-02	6.491029e-02	8.093587e-02	-2.428401e-02
## MEDDM_T4	-5.232131e-02	-9.587091e-03	7.383240e-03	9.461683e-04
## MEDCOL_T4	-5.233223e-02	-8.672076e-03	8.971278e-03	1.229109e-03
## MEDINF_T4	-5.225489e-02	-1.098214e-02	6.806316e-03	3.441130e-03
## MEDHTA_T4	-5.196640e-02	-4.783872e-03	3.883764e-03	9.995554e-03
## GLU_T4	3.776587e-02	9.708867e-03	-5.493337e-03	1.930741e-02
## INS_T4	-8.057575e-03	-1.788429e-02	2.241292e-02	-8.746490e-03
## HOMA_T4	-4.833630e-02	-1.115173e-02	9.878051e-03	3.709219e-03
## HBA1C_T4	-4.735918e-02	-1.419374e-02	2.491052e-02	-2.095342e-03
## HBA1C.mmol.mol_T4	3.326820e-02	2.034290e-03	4.200954e-02	-4.767790e-03
## PESO_T4	3.288495e-02	-8.905267e-03	-2.403978e-02	-4.005446e-02
## bmi_T4	3.094881e-02	-1.048555e-02	-2.612189e-02	-2.319682e-02
## CC_T4	-5.215970e-02	-1.286902e-02	5.244644e-03	-1.417420e-03
## CINT_T4	4.279097e-02	-2.547964e-03	-8.785329e-03	-1.697358e-02
## CAD_T4	3.820527e-02	1.988881e-03	6.380656e-03	-1.054119e-02
## TAD_T4	3.888406e-02	-2.264666e-03	4.187906e-02	7.651832e-03
## TAS_T4	4.239813e-02	5.360805e-03	4.200986e-02	-1.453473e-02
## TG_T4	2.759201e-02	-1.336125e-02	3.672854e-03	5.090264e-02
## COL_T4	3.621061e-02	-8.279696e-03	7.623956e-02	-7.160270e-04
## LDL_T4	3.831689e-02	8.833033e-03	5.682105e-02	1.894601e-02
## HDL_T4	1.453665e-02	-1.994897e-02	6.643885e-02	-2.669485e-02
## VLDL_T4	9.870018e-03	-4.033020e-02	1.802995e-02	5.564629e-02
## PCR_T4	-2.166929e-02	1.217278e-03	2.833520e-02	-2.946358e-02
## LEP_T4	1.455570e-02	-1.568257e-02	2.964047e-02	-6.557642e-03
## ADIPO_T4	-4.504446e-03	1.127687e-02	-3.303355e-02	-2.535030e-02
## GOT_T4	-5.335540e-03	-2.610513e-02	-1.364143e-02	2.050602e-02
## GPT_T4	3.887932e-03	-2.283927e-02	-3.072336e-02	4.906735e-02
## GGT_T4	3.954245e-03	4.308871e-03	1.292712e-02	1.266444e-03
## URICO_T4	-4.156842e-02	-2.498072e-02	2.054556e-02	1.741022e-02
## CREAT_T4	-5.211554e-02	-1.083001e-02	5.202121e-03	-1.823182e-03
## UREA_T4	1.689411e-02	1.411641e-02	-1.529797e-02	3.072658e-02
## HIERRO_T4	4.260482e-03	-2.449553e-02	1.241037e-02	4.234017e-02
## TRANSF_T4	3.678199e-02	-7.980235e-03	6.058152e-02	1.680610e-03
## FERR_T4	1.079934e-02	1.832551e-02	-3.696757e-02	8.470054e-03
## Ile_T4	9.262907e-03	-8.640029e-02	-4.576903e-02	-4.984219e-02
## Leu_T4	9.422412e-03	-8.796663e-02	-5.678239e-02	-7.056775e-02
## Val_T4	1.001374e-02	-8.727427e-02	-5.667147e-02	-8.636508e-02
## Ala_T4	1.726081e-02	-9.022227e-02	-5.553461e-02	-3.481182e-02
## Pro_T4	1.140462e-02	-6.154606e-02	-6.179894e-02	-7.409421e-02

## Gly_T4	5.556011e-03	-7.761374e-02	-3.371671e-03	-7.567848e-02
## Ser_T4	1.463329e-02	-3.311877e-02	-3.267483e-03	-2.855353e-02
## Trp_T4	-1.381675e-03	-9.454777e-02	-3.107555e-02	-8.216936e-02
## Phe_T4	5.788309e-03	-7.585191e-02	-6.444177e-02	-9.993741e-02
## Met_T4	-1.699207e-02	-6.642533e-02	4.589468e-02	-8.793842e-02
## Orn_T4	-2.300997e-03	-8.507586e-02	-3.841970e-02	-7.731444e-02
## Arg_T4	1.727422e-02	-8.035830e-02	-5.833876e-02	-5.863482e-02
## His_T4	7.593302e-03	-8.450097e-02	-7.253802e-02	-7.107802e-02
## Asn_T4	-3.936490e-03	-1.044768e-01	-1.519229e-02	-8.279319e-02
## Asp_T4	-3.282301e-03	-2.824834e-02	-6.080712e-02	-1.565711e-02
## Glu_T4	2.061895e-02	1.539095e-02	-1.058520e-01	5.156662e-02
## Gln_T4	1.303804e-02	-1.050770e-02	5.064413e-02	4.448180e-02
## Cit_T4	-6.288421e-03	-9.584785e-02	-3.430173e-02	-6.596955e-02
## Tyr_T4	1.554605e-03	-8.454654e-02	-4.993335e-02	-8.158786e-02
## Thr_T4	4.983106e-03	-8.900770e-02	2.722186e-03	-7.678495e-02
## Lys_T4	1.420386e-02	-5.292446e-02	1.299039e-02	-2.968199e-02
## Creatinine_T4	7.388703e-03	-8.833154e-02	-5.368672e-02	-5.931448e-02
## Kynurenine_T4	-1.549081e-02	-5.617529e-03	-3.994171e-02	-5.408187e-02
## Putrescine_T4	-4.081949e-03	-1.353443e-02	2.323255e-02	2.264845e-03
## Sarcosine_T4	-2.423799e-02	-3.485859e-02	-8.979389e-03	1.408779e-02
## Serotonin_T4	-1.100873e-02	7.112118e-03	-9.217869e-03	-3.478700e-02
## Taurine_T4	1.222143e-02	3.682083e-02	5.409716e-02	1.939982e-02
## SDMA_T4	-5.133143e-02	-2.778403e-02	3.129862e-04	-1.014630e-02
## C0_T4	-1.577593e-03	-8.812115e-02	-6.511264e-02	-6.111451e-02
## C2_T4	-2.191465e-02	-8.727889e-02	1.952410e-02	-4.392680e-03
## C3.OH_T4	-5.240269e-02	-9.192554e-03	6.613897e-03	-5.927405e-04
## C6..C4.1.DC._T4	3.086552e-03	3.624466e-02	3.066182e-02	3.903211e-02
## C5.DC..C6.OH._T4	-3.776637e-02	2.286194e-03	-2.741385e-02	-3.617937e-02
## C7.DC_T4	-3.918263e-02	-1.357590e-02	-1.525952e-02	1.418613e-03
## C8_T4	-5.231163e-02	-1.044477e-02	6.015725e-03	-2.244538e-03
## C10_T4	-5.228081e-02	-1.251930e-02	4.201056e-03	-2.919374e-03
## C10.1_T4	-5.238288e-02	-1.119416e-02	5.391740e-03	2.614087e-05
## C10.2_T4	-5.242964e-02	-9.721547e-03	7.304200e-03	2.644183e-04
## C14.1_T4	-5.239423e-02	-1.100754e-02	6.403639e-03	-5.615208e-04
## C14.2_T4	-5.242313e-02	-1.008755e-02	6.660763e-03	4.522273e-04
## C16.1_T4	-4.571691e-02	-1.086825e-02	8.704063e-05	6.301137e-03
## C16.2_T4	-5.244948e-02	-9.609665e-03	6.737918e-03	1.135550e-03
## C16.2.OH_T4	-4.582815e-02	-1.059955e-02	1.470787e-05	6.660216e-03
## C18.1_T4	-5.239694e-02	-1.156639e-02	6.212138e-03	-7.840861e-04
## C18.1.OH_T4	-5.244969e-02	-9.540861e-03	6.859970e-03	9.155320e-04
## C18.2_T4	-5.243200e-02	-1.009875e-02	6.686196e-03	5.992285e-04
## lysoPC.a.C16.0_T4	1.713196e-02	-7.138737e-02	-8.209270e-02	-2.393707e-02
## lysoPC.a.C16.1_T4	-4.027309e-02	-4.251549e-02	-5.491598e-02	-2.199679e-02
## lysoPC.a.C17.0_T4	-4.928457e-02	-2.815814e-02	-2.395334e-02	-1.046205e-02
## lysoPC.a.C18.0_T4	1.413599e-02	-5.238706e-02	-8.900639e-02	-1.301527e-02
## lysoPC.a.C18.1_T4	5.038105e-03	-8.082093e-02	-6.025839e-02	-4.679781e-02
## lysoPC.a.C18.2_T4	-1.491493e-03	-9.588730e-02	-5.134648e-02	-4.717478e-02
## lysoPC.a.C20.3_T4	-4.884794e-02	-3.296966e-02	-2.043077e-02	-3.217678e-03
## lysoPC.a.C20.4_T4	-1.417880e-02	-8.274569e-02	-3.412700e-02	-2.518016e-02
## lysoPC.a.C24.0_T4	-5.245141e-02	-8.250638e-03	3.619021e-03	1.066094e-03
## lysoPC.a.C26.0_T4	-3.856085e-02	-1.341375e-02	-1.743713e-02	-1.873343e-02
## lysoPC.a.C26.1_T4	-5.247117e-02	-9.099357e-03	8.160507e-04	8.595927e-04
## lysoPC.a.C28.0_T4	-3.516106e-02	-1.816111e-02	-4.611272e-02	-2.461032e-02
## lysoPC.a.C28.1_T4	-5.246483e-02	-9.063869e-03	-1.807198e-03	3.881281e-03

## PC.aa.C24.0_T4	-3.164774e-02	-9.324753e-03	-2.704195e-02	-1.317167e-02
## PC.aa.C28.1_T4	-5.022019e-02	-3.179324e-02	5.546469e-03	-1.914702e-02
## PC.aa.C30.0_T4	-5.100076e-02	-2.698390e-02	4.912533e-03	-1.292185e-03
## PC.aa.C32.0_T4	-1.647288e-02	-1.003664e-01	-1.015577e-02	-4.807823e-02
## PC.aa.C32.1_T4	-1.892055e-02	-5.558125e-02	-3.809191e-02	-2.862147e-02
## PC.aa.C32.3_T4	-5.245559e-02	-1.173329e-02	5.160321e-03	3.379655e-04
## PC.aa.C34.1_T4	1.409626e-02	-9.104317e-02	-2.358959e-02	-4.504567e-02
## PC.aa.C34.2_T4	7.209987e-03	-1.071653e-01	-1.273804e-02	-3.102972e-02
## PC.aa.C34.3_T4	-2.922942e-02	-6.244900e-02	-3.840478e-02	-1.984401e-02
## PC.aa.C34.4_T4	-5.232296e-02	-1.387201e-02	4.576163e-03	2.359843e-04
## PC.aa.C36.0_T4	-5.173241e-02	-1.899562e-02	-8.811460e-03	2.920794e-03
## PC.aa.C36.1_T4	-1.032037e-02	-7.115917e-02	-1.124918e-02	-3.905434e-02
## PC.aa.C36.2_T4	7.555204e-03	-9.117929e-02	-9.295991e-03	-4.530973e-02
## PC.aa.C36.3_T4	7.099049e-03	-8.297010e-02	-7.683613e-03	-3.580771e-02
## PC.aa.C36.4_T4	7.730246e-03	-9.962392e-02	1.352704e-02	-2.015163e-02
## PC.aa.C36.5_T4	-2.533029e-02	-6.763111e-02	-2.385244e-02	-3.456147e-02
## PC.aa.C38.0_T4	-4.772214e-02	-4.203239e-02	-5.801135e-03	-2.038307e-02
## PC.aa.C38.1_T4	-5.116917e-02	-1.275558e-02	-1.823866e-02	1.873576e-02
## PC.aa.C38.3_T4	5.259916e-03	-7.584427e-02	-9.527220e-03	-3.220045e-02
## PC.aa.C38.4_T4	9.180663e-03	-9.531148e-02	8.052211e-03	-4.808785e-02
## PC.aa.C38.5_T4	-4.965751e-04	-1.000333e-01	-1.741344e-03	-6.204222e-02
## PC.aa.C38.6_T4	6.768952e-03	-8.764110e-02	-3.264173e-03	-7.004607e-03
## PC.aa.C40.1_T4	-3.389054e-02	-2.801690e-02	-3.432404e-02	-4.573889e-03
## PC.aa.C40.2_T4	-5.092157e-02	2.877257e-03	-2.541736e-02	2.067328e-02
## PC.aa.C40.3_T4	-5.171111e-02	-4.423233e-03	-1.904176e-02	1.375604e-02
## PC.aa.C40.4_T4	-4.928070e-02	-1.818413e-02	-3.194470e-02	2.931356e-03
## PC.aa.C40.5_T4	-4.256444e-02	-5.996266e-02	1.345227e-03	-2.742795e-02
## PC.aa.C40.6_T4	-3.830424e-03	-8.943352e-02	-1.282754e-03	-1.147365e-02
## PC.aa.C42.0_T4	-5.226782e-02	-1.641383e-02	-2.238445e-03	-4.074328e-04
## PC.aa.C42.1_T4	-5.249195e-02	-1.077104e-02	-8.519061e-04	2.153613e-03
## PC.aa.C42.2_T4	-5.244790e-02	-6.414860e-03	-7.032941e-03	6.514160e-03
## PC.aa.C42.4_T4	-5.236424e-02	-4.784123e-03	-9.218458e-03	8.616250e-03
## PC.aa.C42.5_T4	-5.249389e-02	-1.162755e-02	-5.653181e-04	3.276734e-03
## PC.aa.C42.6_T4	-4.000615e-02	-2.481959e-02	-2.258466e-02	-2.354775e-03
## PC.ae.C30.0_T4	-4.191499e-02	-2.207880e-02	1.030657e-02	-1.402873e-02
## PC.ae.C32.1_T4	-4.939342e-02	-4.074802e-02	1.649122e-03	-2.386045e-02
## PC.ae.C32.2_T4	-5.218699e-02	-1.831028e-02	1.679541e-03	-6.964678e-03
## PC.ae.C34.0_T4	-5.235198e-02	-1.424114e-02	5.960183e-03	-3.558667e-03
## PC.ae.C34.1_T4	-4.061515e-02	-6.791533e-02	-7.813853e-04	-5.040955e-02
## PC.ae.C34.2_T4	-3.655876e-02	-7.488340e-02	-6.960182e-03	-5.563544e-02
## PC.ae.C34.3_T4	-3.940780e-02	-6.814094e-02	2.714868e-03	-3.598939e-02
## PC.ae.C36.0_T4	-5.216797e-02	-1.831788e-02	1.998333e-04	-5.647126e-03
## PC.ae.C36.1_T4	-2.673550e-02	1.030129e-02	-9.677020e-02	2.719704e-02
## PC.ae.C36.2_T4	-3.968251e-02	-6.131523e-02	-2.554263e-02	-2.504375e-02
## PC.ae.C36.3_T4	-4.817349e-02	-4.603808e-02	-2.356109e-03	-2.583440e-02
## PC.ae.C36.4_T4	-3.365439e-02	-7.733988e-02	5.319832e-07	-6.532096e-02
## PC.ae.C36.5_T4	-3.441805e-02	-8.167917e-02	4.396732e-03	-5.025253e-02
## PC.ae.C38.0_T4	-5.107980e-02	-2.705217e-02	-4.754182e-03	-4.389361e-03
## PC.ae.C38.2_T4	-3.251521e-02	2.003178e-02	-8.543100e-02	3.823899e-02
## PC.ae.C38.3_T4	-3.276376e-02	2.145729e-02	-8.128558e-02	5.303645e-02
## PC.ae.C38.4_T4	-4.280554e-02	-5.827199e-02	-5.359582e-03	-3.899559e-02
## PC.ae.C38.5_T4	-2.607104e-02	-8.351648e-02	8.783486e-03	-6.756581e-02
## PC.ae.C38.6_T4	-4.580786e-02	-5.588684e-02	5.973495e-03	-3.165104e-02
## PC.ae.C40.1_T4	-5.151444e-02	-1.217382e-02	-2.108544e-02	1.003652e-02

## PC.ae.C40.2_T4	-5.039888e-02	-7.600454e-03	-2.773796e-02	1.385474e-02
## PC.ae.C40.3_T4	-3.745867e-02	1.955701e-02	-7.309303e-02	4.691796e-02
## PC.ae.C40.4_T4	-4.579107e-02	-5.475700e-03	-5.314290e-02	1.583794e-02
## PC.ae.C40.5_T4	-4.060167e-02	-1.832017e-03	-7.349792e-02	1.551030e-02
## PC.ae.C40.6_T4	-5.003971e-02	-3.815300e-02	-1.241504e-03	-1.623764e-02
## PC.ae.C42.1_T4	-5.215109e-02	-5.778763e-03	-1.236943e-02	1.216057e-02
## PC.ae.C42.2_T4	-5.232789e-02	-7.731871e-03	-9.310328e-03	7.574398e-03
## PC.ae.C42.3_T4	-5.168568e-02	-8.194570e-03	-2.183769e-02	8.086212e-03
## PC.ae.C42.4_T4	-5.212115e-02	-1.336107e-02	-8.742491e-03	6.003850e-04
## PC.ae.C42.5_T4	-4.821479e-02	-3.021840e-02	-3.149769e-02	-1.430080e-02
## PC.ae.C44.3_T4	-5.253276e-02	-6.925788e-03	-2.523453e-03	4.897807e-03
## PC.ae.C44.4_T4	-5.243211e-02	-1.157038e-02	2.064830e-03	8.127635e-04
## PC.ae.C44.5_T4	-4.808206e-02	-4.321302e-02	-2.101124e-03	-2.560206e-02
## PC.ae.C44.6_T4	-5.065478e-02	-3.031522e-02	4.643480e-04	-1.293130e-02
## SM.OH.C14.1_T4	-4.464298e-02	-5.221134e-02	8.126572e-03	-3.452157e-02
## SM.OH.C16.1_T4	-5.067272e-02	-2.882120e-02	9.079541e-03	-8.134989e-03
## SM.OH.C22.1_T4	-4.611094e-02	-4.176324e-02	1.554981e-02	-1.572935e-02
## SM.OH.C22.2_T4	-4.448851e-02	-5.348386e-02	-7.229770e-03	-1.426901e-02
## SM.OH.C24.1_T4	-5.228478e-02	-1.415748e-02	6.742601e-03	-1.739562e-04
## SM.C16.0_T4	1.291397e-02	-1.018592e-01	6.458077e-03	-4.162183e-02
## SM.C16.1_T4	-1.581946e-02	-9.631158e-02	-7.437753e-03	-4.231515e-02
## SM.C18.0_T4	-4.589461e-03	-8.799884e-02	8.513275e-03	1.926188e-02
## SM.C18.1_T4	-2.294203e-02	-8.631346e-02	-3.427442e-03	-1.092838e-02
## SM.C20.2_T4	-5.237725e-02	-1.306412e-02	6.092465e-03	6.583696e-04
## SM.C24.0_T4	-3.860105e-02	-5.500487e-02	-2.059113e-03	-1.005620e-02
## SM.C24.1_T4	1.019777e-02	-9.989735e-02	-7.273003e-03	-6.633316e-03
## MEDDM_T5	-5.150080e-02	-3.237145e-03	3.624385e-03	1.010204e-02
## MEDCOL_T5	-5.150080e-02	-3.237145e-03	3.624385e-03	1.010204e-02
## MEDINF_T5	-5.147425e-02	-4.520952e-03	3.540330e-03	1.237918e-02
## MEDHTA_T5	-5.144547e-02	-3.290546e-03	4.415548e-03	9.942508e-03
## GLU_T5	-4.932356e-03	8.827364e-02	2.392928e-02	2.086465e-02
## INS_T5	-3.656599e-02	3.097675e-02	1.651235e-02	3.040778e-02
## HOMA_T5	-5.044671e-02	5.744586e-03	1.377589e-02	1.311854e-02
## HBA1C_T5	-4.740661e-02	-1.726258e-02	2.114776e-02	-1.609879e-03
## HBA1C.mmol.mol_T5	3.435251e-02	-8.243280e-03	2.939061e-02	-3.154598e-03
## PESO_T5	-2.080252e-02	8.848088e-02	-3.398739e-02	-6.389771e-03
## bmi_T5	-2.237296e-02	9.031373e-02	-3.361876e-02	6.714940e-03
## CC_T5	-5.219480e-02	-1.130866e-02	8.353558e-03	1.394497e-03
## CINT_T5	-2.298862e-02	9.986742e-02	-2.057489e-02	2.146791e-02
## CAD_T5	-1.107638e-02	1.027230e-01	-2.940514e-02	2.045903e-03
## TAD_T5	3.879234e-02	5.728967e-03	2.662768e-02	2.104849e-02
## TAS_T5	4.134639e-02	1.224887e-02	1.332890e-02	2.122096e-03
## TG_T5	-1.296414e-02	5.106110e-02	1.744704e-02	6.915014e-02
## COL_T5	1.030419e-02	3.538364e-02	9.360254e-02	2.797949e-02
## LDL_T5	2.068921e-02	4.370577e-02	6.554792e-02	3.320838e-02
## HDL_T5	-9.409149e-03	7.674480e-03	6.183629e-02	1.216976e-02
## VLDL_T5	-3.165450e-02	3.064036e-02	1.568542e-02	7.712265e-02
## PCR_T5	-3.820377e-02	2.402352e-02	3.775250e-02	-4.537154e-02
## LEP_T5	-2.092006e-02	7.537560e-02	2.583980e-02	-1.449497e-02
## ADIPO_T5	-1.194825e-02	2.956043e-02	-1.952951e-02	-9.008070e-03
## GOT_T5	-2.025115e-02	3.161952e-02	-7.807932e-03	8.795260e-03
## GPT_T5	-1.211157e-02	5.657011e-02	-2.766752e-02	2.687791e-02
## GGT_T5	-3.310877e-02	7.348772e-02	-1.661329e-02	4.534138e-02
## URICO_T5	-4.899134e-02	1.976455e-02	2.344549e-04	2.449636e-02

## CREAT_T5	-5.247049e-02	-4.689248e-03	6.840599e-03	-1.884846e-04
## UREA_T5	-1.447445e-02	6.533065e-02	-1.604312e-02	4.974307e-03
## HIERRO_T5	-4.796012e-03	6.332439e-04	1.148873e-02	7.421884e-02
## TRANSF_T5	-1.355665e-02	6.965061e-02	3.776407e-02	2.406428e-02
## FERR_T5	1.195663e-03	2.637314e-02	-4.554595e-02	3.085219e-02
## Ile_T5	3.465359e-02	-8.079058e-03	-2.817134e-02	5.863541e-02
## Leu_T5	3.353699e-02	-1.673166e-02	-1.775763e-02	4.551440e-02
## Val_T5	3.577992e-02	-1.360018e-02	3.709242e-03	3.393547e-02
## Ala_T5	4.356605e-02	1.599638e-02	-1.211484e-02	6.537560e-02
## Pro_T5	3.821640e-02	-8.080291e-03	-1.398297e-02	2.737739e-02
## Gly_T5	2.963037e-02	-2.137027e-02	3.944158e-02	3.703296e-02
## Ser_T5	3.513482e-02	-3.761597e-02	3.179049e-02	1.723059e-02
## Trp_T5	2.121909e-02	-1.242593e-02	4.457891e-02	5.339619e-02
## Phe_T5	3.252321e-02	-2.805376e-03	-1.502318e-02	4.471480e-02
## Met_T5	1.384154e-02	-4.230277e-02	8.062186e-02	2.833562e-02
## Orn_T5	2.846596e-02	-2.453541e-02	2.363622e-02	4.225131e-02
## Arg_T5	3.627374e-02	-3.391671e-04	-2.278124e-03	9.847268e-02
## His_T5	3.298185e-02	-8.154079e-03	-4.140090e-02	7.439198e-02
## Asn_T5	3.107940e-02	-3.627083e-02	4.541242e-02	4.194208e-02
## Asp_T5	9.369929e-03	1.453723e-02	2.249959e-02	4.561277e-02
## Glu_T5	2.226301e-02	4.292080e-02	-7.259913e-02	6.952453e-02
## Gln_T5	2.810448e-02	-5.990639e-02	5.466750e-02	4.683917e-02
## Cit_T5	2.621963e-02	-7.355723e-03	2.464372e-02	6.534109e-02
## Tyr_T5	3.543152e-02	9.604781e-04	8.252083e-03	5.498797e-02
## Thr_T5	3.183640e-02	-1.746851e-02	7.279472e-02	1.029979e-02
## Lys_T5	4.125049e-02	-6.561489e-03	1.465518e-02	6.663065e-02
## Creatinine_T5	2.401439e-02	-2.828510e-02	3.363679e-02	3.403529e-02
## Kynurenine_T5	-6.222454e-03	-8.007623e-03	9.589713e-02	1.346008e-02
## Putrescine_T5	-9.185678e-03	-3.704751e-03	-3.769871e-02	4.288502e-02
## Sarcosine_T5	-2.136796e-02	1.621620e-02	3.700191e-03	3.974124e-02
## Serotonin_T5	-1.326178e-02	1.282109e-03	-1.678207e-02	-3.395655e-02
## Taurine_T5	2.022184e-02	1.322162e-02	1.453510e-02	-3.155788e-02
## SDMA_T5	-5.184739e-02	-1.465096e-02	1.452524e-02	1.157047e-02
## C0_T5	2.960476e-02	4.178836e-03	2.111815e-03	8.299266e-02
## C2_T5	-2.218936e-02	-4.967117e-02	6.677972e-02	4.401575e-02
## C3.OH_T5	-5.238162e-02	-1.044421e-02	7.088899e-03	8.029096e-05
## C6..C4.1.DC._T5	-3.334221e-03	1.183923e-03	-4.458075e-02	3.625662e-02
## C5.DC..C6.OH._T5	-4.574370e-02	-6.148936e-03	-3.807181e-03	-1.703594e-02
## C7.DC_T5	-4.120378e-02	-2.786588e-02	2.621333e-02	-2.092243e-02
## C8_T5	-5.225193e-02	-1.067114e-02	8.435645e-03	8.035746e-05
## C10_T5	-5.216199e-02	-1.137957e-02	8.743399e-03	1.517650e-03
## C10.1_T5	-4.000612e-02	-1.197213e-02	1.832955e-02	3.550731e-02
## C10.2_T5	-5.239715e-02	-1.026692e-02	7.127743e-03	6.733821e-04
## C14.1_T5	-5.238913e-02	-1.048270e-02	7.702819e-03	1.966301e-03
## C14.2_T5	-5.241707e-02	-9.923341e-03	6.784646e-03	1.245400e-03
## C16.1_T5	-5.243229e-02	-9.892061e-03	6.872788e-03	1.296953e-03
## C16.2_T5	-5.244414e-02	-9.750169e-03	6.708102e-03	1.064722e-03
## C16.2.OH_T5	-4.268832e-02	-1.280140e-02	-3.344134e-02	-3.205222e-02
## C18.1_T5	-5.242291e-02	-1.004155e-02	7.210269e-03	2.320689e-03
## C18.1.OH_T5	-5.244673e-02	-9.784750e-03	6.796257e-03	1.004841e-03
## C18.2_T5	-5.243392e-02	-9.888021e-03	6.550968e-03	1.622239e-03
## lysoPC.a.C16.0_T5	2.907325e-02	-1.432611e-02	-4.024542e-03	1.049269e-01
## lysoPC.a.C16.1_T5	-3.770926e-02	-3.027073e-03	-3.943408e-02	5.920804e-02
## lysoPC.a.C17.0_T5	-4.988898e-02	-1.018307e-02	1.042120e-02	3.926384e-02

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## lysoPC.a.C18.0_T5 2.688807e-02 6.407240e-03 -4.838211e-03 1.145931e-01
## lysoPC.a.C18.1_T5 2.054466e-02 -2.087372e-02 2.111059e-03 9.161491e-02
## lysoPC.a.C18.2_T5 1.494636e-02 -3.181746e-02 1.523820e-02 1.036081e-01
## lysoPC.a.C20.3_T5 -4.278276e-02 -1.101158e-02 -2.985152e-03 7.534093e-02
## lysoPC.a.C20.4_T5 -1.554890e-03 -2.765956e-02 9.501379e-03 9.692562e-02
## lysoPC.a.C24.0_T5 -5.240408e-02 -8.990629e-03 5.033681e-03 3.108793e-03
## lysoPC.a.C26.0_T5 -5.242397e-02 -6.845549e-03 -4.610364e-04 8.050022e-03
## lysoPC.a.C26.1_T5 -5.240065e-02 -1.021802e-02 5.164947e-03 5.007766e-03
## lysoPC.a.C28.0_T5 -3.058255e-02 -8.995770e-03 4.522804e-03 7.876446e-02
## lysoPC.a.C28.1_T5 -5.243372e-02 -9.282513e-03 4.969562e-03 6.307637e-03
## PC.aa.C24.0_T5 -2.996863e-02 -4.164895e-03 1.888470e-02 1.128320e-02
## PC.aa.C28.1_T5 -5.049343e-02 -2.076042e-02 2.944869e-02 2.385457e-02
## PC.aa.C30.0_T5 -5.022501e-02 -2.013622e-02 1.900439e-02 2.462978e-02
## PC.aa.C32.0_T5 -1.299253e-02 -6.246764e-02 8.907183e-02 9.105592e-02
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## PC.aa.C32.3_T5 -5.244319e-02 -9.996895e-03 7.662508e-03 2.925786e-03
## PC.aa.C34.1_T5 2.626829e-02 -4.407915e-02 7.721770e-02 9.794823e-02
## PC.aa.C34.2_T5 2.013604e-02 -5.430590e-02 8.325655e-02 1.031700e-01
## PC.aa.C34.3_T5 -2.049242e-02 -3.036288e-02 3.911787e-02 1.015323e-01
## PC.aa.C34.4_T5 -5.226023e-02 -1.091114e-02 7.744382e-03 8.979559e-03
## PC.aa.C36.0_T5 -5.154809e-02 -1.118056e-02 3.369201e-03 2.459519e-02
## PC.aa.C36.1_T5 4.658680e-03 -2.188069e-02 7.480618e-02 1.242558e-01
## PC.aa.C36.2_T5 2.214578e-02 -3.266877e-02 8.478424e-02 1.080345e-01
## PC.aa.C36.3_T5 2.272746e-02 -2.994081e-02 7.403130e-02 1.099153e-01
## PC.aa.C36.4_T5 2.079923e-02 -5.039248e-02 7.343672e-02 8.974764e-02
## PC.aa.C36.5_T5 -2.826087e-02 -2.924563e-02 4.671139e-02 7.374215e-02
## PC.aa.C38.0_T5 -4.741980e-02 -2.308284e-02 2.709394e-02 4.359518e-02
## PC.aa.C38.1_T5 -5.146515e-02 -2.945281e-03 5.442573e-03 1.679289e-02
## PC.aa.C38.3_T5 1.901366e-02 -1.173850e-02 4.177319e-02 1.178720e-01
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## PC.aa.C38.5_T5 1.667028e-02 -4.590749e-02 9.941096e-02 8.733492e-02
## PC.aa.C38.6_T5 2.250060e-02 -3.984975e-02 6.314098e-02 1.135098e-01
## PC.aa.C40.1_T5 -3.198230e-02 -1.566270e-02 1.408036e-02 5.864319e-02
## PC.aa.C40.2_T5 -5.171586e-02 -8.661961e-04 -1.559623e-02 1.466316e-02
## PC.aa.C40.3_T5 -5.169293e-02 -3.568410e-03 -9.416695e-03 1.329640e-02
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## PC.aa.C42.0_T5 -5.231470e-02 -1.156519e-02 6.669088e-03 9.862692e-03
## PC.aa.C42.1_T5 -5.246941e-02 -9.591139e-03 4.616755e-03 5.910429e-03
## PC.aa.C42.2_T5 -5.244764e-02 -6.501827e-03 -2.657050e-03 7.038771e-03
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## PC.aa.C42.6_T5 -3.883784e-02 4.446195e-03 2.885925e-02 2.931491e-02
## PC.ae.C30.0_T5 -4.338125e-02 1.230531e-03 3.757069e-02 1.544726e-02
## PC.ae.C32.1_T5 -5.000798e-02 -2.463406e-02 3.016962e-02 2.190687e-02
## PC.ae.C32.2_T5 -5.228321e-02 -1.246613e-02 1.090434e-02 6.976597e-03
## PC.ae.C34.0_T5 -5.230840e-02 -1.059795e-02 1.208706e-02 5.232805e-03
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## PC.ae.C34.2_T5 -3.249940e-02 -4.658514e-02 7.552751e-02 6.816774e-02
## PC.ae.C34.3_T5 -3.650278e-02 -4.556629e-02 5.977278e-02 5.952175e-02
## PC.ae.C36.0_T5 -5.214814e-02 -1.253209e-02 1.143110e-02 1.205451e-02
## PC.ae.C36.1_T5 -3.399604e-02 1.780170e-02 -4.805272e-02 6.258786e-02
## PC.ae.C36.2_T5 -3.698098e-02 -2.696795e-02 5.992963e-02 7.752413e-02

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## PC.ae.C36.3_T5	-4.709058e-02	-2.744909e-02	4.481171e-02	3.842484e-02
## PC.ae.C36.4_T5	-3.082485e-02	-5.476116e-02	7.887616e-02	5.879241e-02
## PC.ae.C36.5_T5	-2.920571e-02	-5.754691e-02	7.636102e-02	6.537629e-02
## PC.ae.C38.0_T5	-5.165695e-02	-1.421318e-02	1.269032e-02	2.363743e-02
## PC.ae.C38.2_T5	-3.716498e-02	1.788185e-02	-5.784289e-02	4.351079e-02
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## PC.ae.C38.4_T5	-4.211157e-02	-2.866996e-02	6.975017e-02	5.656605e-02
## PC.ae.C38.5_T5	-1.797039e-02	-5.504358e-02	9.519102e-02	6.293168e-02
## PC.ae.C38.6_T5	-4.478431e-02	-3.529525e-02	5.314466e-02	4.865804e-02
## PC.ae.C40.1_T5	-5.186014e-02	-7.165665e-03	-3.167751e-03	2.450672e-02
## PC.ae.C40.2_T5	-5.085067e-02	-1.608866e-03	-1.202225e-02	3.121778e-02
## PC.ae.C40.3_T5	-4.110564e-02	2.082306e-02	-5.621954e-02	5.430364e-02
## PC.ae.C40.4_T5	-4.847784e-02	2.352035e-03	-1.905042e-02	4.171745e-02
## PC.ae.C40.5_T5	-4.479973e-02	5.846900e-03	-3.249299e-02	5.523492e-02
## PC.ae.C40.6_T5	-5.007202e-02	-2.001413e-02	3.049148e-02	3.288928e-02
## PC.ae.C42.1_T5	-5.233277e-02	-5.129929e-03	-5.171104e-03	1.303470e-02
## PC.ae.C42.2_T5	-5.240686e-02	-5.697950e-03	-1.342449e-03	1.010146e-02
## PC.ae.C42.3_T5	-5.199001e-02	-5.210296e-03	-7.307275e-03	1.701921e-02
## PC.ae.C42.4_T5	-5.214477e-02	-9.002769e-03	2.398404e-03	1.525033e-02
## PC.ae.C42.5_T5	-4.919234e-02	-1.100229e-02	-5.061407e-03	4.178787e-02
## PC.ae.C44.3_T5	-5.249463e-02	-6.823690e-03	-1.539056e-03	6.112403e-03
## PC.ae.C44.4_T5	-5.240055e-02	-9.800095e-03	6.272282e-03	6.023474e-03
## PC.ae.C44.5_T5	-4.952147e-02	-2.504813e-02	2.919331e-02	2.260792e-02
## PC.ae.C44.6_T5	-5.081071e-02	-2.120197e-02	1.897629e-02	1.626352e-02
## SM..OH..C14.1_T5	-4.546559e-02	-2.704088e-02	6.026883e-02	3.922726e-02
## SM..OH..C16.1_T5	-5.123293e-02	-1.555209e-02	2.719542e-02	1.767007e-02
## SM..OH..C22.1_T5	-4.081127e-02	-3.174242e-02	5.776840e-02	4.910790e-02
## SM..OH..C22.2_T5	-4.280667e-02	-2.744135e-02	4.966870e-02	6.182782e-02
## SM..OH..C24.1_T5	-5.225125e-02	-1.193210e-02	1.215401e-02	4.825664e-03
## SM.C16.0_T5	2.511732e-02	-4.307651e-02	9.079469e-02	9.336900e-02
## SM.C16.1_T5	-1.059342e-02	-5.039223e-02	9.530042e-02	7.761307e-02
## SM.C18.0_T5	1.577563e-03	-3.445486e-02	7.254780e-02	1.129948e-01
## SM.C18.1_T5	-2.851238e-02	-3.932298e-02	6.635420e-02	7.695362e-02
## SM.C20.2_T5	-5.234350e-02	-1.104973e-02	8.486420e-03	4.303566e-03
## SM.C24.0_T5	-2.543839e-02	-4.487573e-02	4.899507e-02	9.971798e-02
## SM.C24.1_T5	1.987643e-02	-4.300291e-02	6.967238e-02	1.136367e-01
##	PC5			
## GLU_T0	-1.720638e-02			
## INS_T0	-4.680565e-03			
## HOMA_T0	-3.741193e-03			
## HBA1C_T0	-3.651642e-03			
## HBA1C.mmol.mol_T0	-7.868349e-03			
## PES0_T0	-1.571945e-02			
## bmi_T0	-1.499417e-02			
## CC_T0	-8.783815e-04			
## CINT_T0	-9.147539e-03			
## CAD_T0	-2.201109e-02			
## TAD_T0	1.245867e-03			
## TAS_T0	-2.246549e-03			
## TG_T0	1.198507e-03			
## COL_T0	-5.145790e-02			
## LDL_T0	-5.645871e-02			
## HDL_T0	9.606825e-03			
## VLDL_T0	1.300810e-03			

## PCR_T0	-6.154786e-02
## LEP_T0	-2.436914e-02
## ADIPO_T0	5.368261e-02
## GOT_T0	6.108101e-02
## GPT_T0	2.777123e-02
## GGT_T0	4.267764e-03
## URICO_T0	-5.514437e-03
## CREAT_T0	-2.221790e-03
## UREA_T0	2.230292e-02
## HIERRO_T0	-1.598124e-02
## TRANSF_T0	-2.004095e-02
## FERR_T0	1.554729e-02
## Ile_T0	-5.137047e-02
## Leu_T0	-4.979559e-02
## Val_T0	-8.624054e-02
## Ala_T0	-5.731886e-02
## Pro_T0	-5.170318e-02
## Gly_T0	2.584130e-02
## Ser_T0	9.667183e-03
## Trp_T0	-7.514744e-02
## Phe_T0	-3.174623e-03
## Met_T0	-4.561577e-02
## Orn_T0	2.626557e-02
## Arg_T0	3.758529e-03
## His_T0	-5.772913e-02
## Asn_T0	-7.118079e-02
## Asp_T0	5.081524e-02
## Glu_T0	2.911597e-02
## Gln_T0	-8.858740e-02
## Cit_T0	-3.348022e-02
## Tyr_T0	-3.299038e-02
## Thr_T0	-3.758933e-02
## Lys_T0	-8.360483e-02
## Creatinine_T0	-7.268980e-02
## Kynurenine_T0	-2.184854e-02
## Putrescine_T0	-1.422032e-03
## Sarcosine_T0	-4.702161e-02
## Serotonin_T0	-3.660384e-02
## Taurine_T0	2.303257e-02
## SDMA_T0	-1.083996e-02
## CO_T0	-5.546852e-02
## C2_T0	-5.357889e-02
## C3.OH_T0	-8.049950e-04
## C6..C4.1.DC._T0	6.345401e-02
## C5.DC..C6.OH._T0	7.444036e-03
## C7.DC_T0	2.030726e-02
## C8_T0	5.195643e-04
## C10_T0	-5.194313e-03
## C10.1_T0	-1.149999e-03
## C10.2_T0	-1.097363e-03
## C14.1_T0	-1.253263e-03
## C14.2_T0	-3.964059e-04
## C16.1_T0	-4.211973e-04
## C16.2_T0	1.722988e-04

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## C16.2.OH_T0      3.835253e-04
## C18.1_T0         -5.738362e-04
## C18.1.OH_T0      3.341911e-04
## C18.2_T0         -8.515310e-05
## lysoPC.a.C16.0_T0 -1.995276e-02
## lysoPC.a.C16.1_T0 -9.343917e-03
## lysoPC.a.C17.0_T0  3.778171e-04
## lysoPC.a.C18.0_T0 -4.834820e-03
## lysoPC.a.C18.1_T0 -4.225782e-02
## lysoPC.a.C18.2_T0 -5.465281e-02
## lysoPC.a.C20.3_T0 -3.637022e-02
## lysoPC.a.C20.4_T0 -6.922633e-02
## lysoPC.a.C24.0_T0 -1.561510e-03
## lysoPC.a.C26.0_T0 -1.616745e-02
## lysoPC.a.C26.1_T0 -1.729590e-03
## lysoPC.a.C28.0_T0  4.752687e-03
## lysoPC.a.C28.1_T0 -1.893440e-03
## PC.aa.C24.0_T0   -7.983825e-03
## PC.aa.C28.1_T0   -3.280171e-02
## PC.aa.C30.0_T0   -2.399085e-02
## PC.aa.C32.0_T0   -8.762208e-02
## PC.aa.C32.1_T0   -7.973980e-02
## PC.aa.C32.3_T0   -1.634294e-03
## PC.aa.C34.1_T0   -1.358691e-01
## PC.aa.C34.2_T0   -1.123287e-01
## PC.aa.C34.3_T0   -7.558684e-02
## PC.aa.C34.4_T0   -1.114983e-02
## PC.aa.C36.0_T0   -4.866453e-03
## PC.aa.C36.1_T0   -8.995863e-02
## PC.aa.C36.2_T0   -1.065939e-01
## PC.aa.C36.3_T0   -1.181771e-01
## PC.aa.C36.4_T0   -1.507633e-01
## PC.aa.C36.5_T0   -9.318970e-02
## PC.aa.C38.0_T0   -2.628371e-02
## PC.aa.C38.1_T0   4.536433e-02
## PC.aa.C38.3_T0   -1.040235e-01
## PC.aa.C38.4_T0   -1.375978e-01
## PC.aa.C38.5_T0   -1.360513e-01
## PC.aa.C38.6_T0   -1.293613e-01
## PC.aa.C40.1_T0   -1.007534e-02
## PC.aa.C40.2_T0    2.054555e-02
## PC.aa.C40.3_T0    1.201239e-02
## PC.aa.C40.4_T0   -7.560109e-03
## PC.aa.C40.5_T0   -6.222018e-02
## PC.aa.C40.6_T0   -1.213666e-01
## PC.aa.C42.0_T0   -3.384753e-04
## PC.aa.C42.1_T0    1.536492e-03
## PC.aa.C42.2_T0    6.728100e-03
## PC.aa.C42.4_T0    1.079405e-02
## PC.aa.C42.5_T0    2.582920e-03
## PC.aa.C42.6_T0   -9.373045e-03
## PC.ae.C30.0_T0   -1.338956e-03
## PC.ae.C32.1_T0   -2.601108e-02
## PC.ae.C32.2_T0   -4.140936e-03

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## PC.ae.C34.0_T0      -4.019683e-03
## PC.ae.C34.1_T0      -5.235666e-02
## PC.ae.C34.2_T0      -5.706711e-02
## PC.ae.C34.3_T0      -4.383190e-02
## PC.ae.C36.0_T0      -5.235953e-03
## PC.ae.C36.1_T0       1.866281e-02
## PC.ae.C36.2_T0      -2.318909e-02
## PC.ae.C36.3_T0      -4.434208e-02
## PC.ae.C36.4_T0      -8.882630e-02
## PC.ae.C36.5_T0      -8.686783e-02
## PC.ae.C38.0_T0      -2.114916e-02
## PC.ae.C38.2_T0       4.363949e-02
## PC.ae.C38.3_T0       2.251487e-02
## PC.ae.C38.4_T0      -5.871994e-02
## PC.ae.C38.5_T0      -1.054947e-01
## PC.ae.C38.6_T0      -6.775473e-02
## PC.ae.C40.1_T0       1.541510e-03
## PC.ae.C40.2_T0       2.551408e-02
## PC.ae.C40.3_T0       3.878820e-02
## PC.ae.C40.4_T0       1.397030e-02
## PC.ae.C40.5_T0       1.469480e-02
## PC.ae.C40.6_T0      -2.906283e-02
## PC.ae.C42.1_T0       9.406807e-03
## PC.ae.C42.2_T0       5.632220e-03
## PC.ae.C42.3_T0       9.635461e-03
## PC.ae.C42.4_T0       1.946517e-03
## PC.ae.C42.5_T0       7.755425e-04
## PC.ae.C44.3_T0       7.631846e-03
## PC.ae.C44.4_T0       1.008239e-03
## PC.ae.C44.5_T0      -2.071341e-02
## PC.ae.C44.6_T0      -9.501347e-03
## SM..OH..C14.1_T0    -3.141344e-02
## SM..OH..C16.1_T0    -1.588592e-02
## SM..OH..C22.1_T0    -9.104943e-02
## SM..OH..C22.2_T0    -5.178924e-02
## SM..OH..C24.1_T0    -8.649874e-03
## SM.C16.0_T0          -1.488562e-01
## SM.C16.1_T0          -1.130297e-01
## SM.C18.0_T0          -9.126795e-02
## SM.C18.1_T0          -7.861392e-02
## SM.C20.2_T0          -2.986033e-03
## SM.C24.0_T0          -1.067910e-01
## SM.C24.1_T0          -1.317757e-01
## MEDDM_T2             5.335230e-04
## MEDCOL_T2            5.335230e-04
## MEDINF_T2            -1.621857e-03
## MEDHTA_T2            -2.083133e-04
## GLU_T2               -1.118887e-02
## INS_T2                3.090819e-02
## HOMA_T2               1.440253e-02
## HBA1C_T2             -2.599691e-04
## HBA1C.mmol.mol_T2    2.548893e-03
## PESO_T2              -8.422618e-03
## bmi_T2               -7.553918e-03

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## CC_T2	5.353340e-04
## CINT_T2	-3.628044e-03
## CAD_T2	-1.030943e-02
## TAD_T2	1.854242e-02
## TAS_T2	-8.975238e-03
## TG_T2	-8.086568e-03
## COL_T2	-3.390887e-02
## LDL_T2	-3.529295e-02
## HDL_T2	-1.236031e-02
## VLDL_T2	1.734216e-02
## PCR_T2	-5.048295e-02
## LEP_T2	-2.613449e-02
## ADIPO_T2	5.953662e-02
## GOT_T2	7.922514e-02
## GPT_T2	2.797536e-02
## GGT_T2	4.093309e-03
## URICO_T2	6.449549e-03
## CREAT_T2	-1.101451e-03
## UREA_T2	9.246213e-03
## HIERRO_T2	2.519406e-02
## TRANSF_T2	-5.450842e-03
## FERR_T2	2.388398e-02
## Ile_T2	1.721587e-02
## Leu_T2	4.333834e-02
## Val_T2	3.157222e-02
## Ala_T2	2.511295e-02
## Pro_T2	2.682571e-02
## Gly_T2	6.015683e-02
## Ser_T2	5.469659e-02
## Trp_T2	1.051456e-02
## Phe_T2	5.919895e-02
## Met_T2	-4.358204e-02
## Orn_T2	8.910752e-02
## Arg_T2	6.449982e-02
## His_T2	5.598669e-02
## Asn_T2	2.967764e-02
## Asp_T2	9.303459e-02
## Glu_T2	8.972179e-02
## Gln_T2	1.979045e-03
## Cit_T2	5.104910e-02
## Tyr_T2	3.225136e-02
## Thr_T2	1.405572e-02
## Lys_T2	3.625433e-02
## Creatinine_T2	1.057030e-02
## Kynurenine_T2	4.286078e-03
## Putrescine_T2	-3.517004e-02
## Sarcosine_T2	-1.389908e-02
## Serotonin_T2	2.677595e-03
## Taurine_T2	-1.135532e-02
## SDMA_T2	2.331070e-03
## CO_T2	5.136815e-02
## C2_T2	-2.619780e-02
## C3.OH_T2	-3.210998e-04
## C6..C4.1.DC._T2	-4.799601e-02

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## C5.DC..C6.OH._T2    7.977267e-03
## C7.DC_T2             -6.199207e-03
## C8_T2                -2.038211e-04
## C10_T2               2.576608e-04
## C10.1_T2             8.993374e-04
## C10.2_T2             -1.168486e-03
## C14.1_T2             8.301354e-04
## C14.2_T2             3.900950e-04
## C16.1_T2             3.981869e-04
## C16.2_T2             1.703462e-04
## C16.2.OH_T2          4.198663e-04
## C18.1_T2             9.965502e-04
## C18.1.OH_T2          3.563761e-04
## C18.2_T2             5.830794e-04
## lysoPC.a.C14.0_T2    4.129155e-02
## lysoPC.a.C16.0_T2    4.867326e-02
## lysoPC.a.C16.1_T2    2.112456e-02
## lysoPC.a.C17.0_T2    1.870546e-02
## lysoPC.a.C18.0_T2    4.008249e-02
## lysoPC.a.C18.1_T2    3.494736e-02
## lysoPC.a.C18.2_T2    2.967963e-02
## lysoPC.a.C20.3_T2    1.679209e-02
## lysoPC.a.C20.4_T2    2.233838e-02
## lysoPC.a.C24.0_T2    2.521468e-03
## lysoPC.a.C26.0_T2    2.314272e-02
## lysoPC.a.C26.1_T2    3.669205e-03
## lysoPC.a.C28.0_T2    7.017750e-02
## lysoPC.a.C28.1_T2    7.107187e-03
## PC.aa.C24.0_T2        4.792009e-02
## PC.aa.C28.1_T2        -2.258037e-03
## PC.aa.C30.0_T2        -2.257284e-03
## PC.aa.C32.0_T2        -1.661137e-02
## PC.aa.C32.1_T2        -1.024004e-02
## PC.aa.C32.3_T2        1.468439e-03
## PC.aa.C34.1_T2        -2.412064e-02
## PC.aa.C34.2_T2        -2.398395e-02
## PC.aa.C34.3_T2        4.525159e-03
## PC.aa.C34.4_T2        4.276433e-04
## PC.aa.C36.0_T2        1.408847e-02
## PC.aa.C36.1_T2        -1.320246e-02
## PC.aa.C36.2_T2        -1.097672e-02
## PC.aa.C36.3_T2        -7.883061e-03
## PC.aa.C36.4_T2        -6.117462e-02
## PC.aa.C36.5_T2        -1.562816e-02
## PC.aa.C38.0_T2        3.405639e-04
## PC.aa.C38.1_T2        3.208236e-02
## PC.aa.C38.3_T2        -1.969331e-03
## PC.aa.C38.4_T2        -3.046920e-02
## PC.aa.C38.5_T2        -2.194054e-02
## PC.aa.C38.6_T2        -5.865667e-02
## PC.aa.C40.1_T2        5.417858e-02
## PC.aa.C40.2_T2        3.038402e-02
## PC.aa.C40.3_T2        2.418455e-02
## PC.aa.C40.4_T2        2.630906e-02

```

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## PC.aa.C40.5_T2      1.564386e-03
## PC.aa.C40.6_T2     -3.603300e-02
## PC.aa.C42.0_T2      7.115705e-03
## PC.aa.C42.1_T2      6.158132e-03
## PC.aa.C42.2_T2      1.484610e-02
## PC.aa.C42.4_T2      1.320982e-02
## PC.aa.C42.5_T2      6.680523e-03
## PC.aa.C42.6_T2      1.996814e-02
## PC.ae.C30.0_T2      3.344044e-02
## PC.ae.C32.1_T2     -1.816300e-03
## PC.ae.C32.2_T2      2.381175e-03
## PC.ae.C34.0_T2      5.582478e-04
## PC.ae.C34.1_T2     -2.985953e-04
## PC.ae.C34.2_T2      5.129224e-04
## PC.ae.C34.3_T2     -1.300180e-03
## PC.ae.C36.0_T2      4.867268e-03
## PC.ae.C36.1_T2      7.044954e-02
## PC.ae.C36.2_T2      1.700228e-02
## PC.ae.C36.3_T2     -1.330701e-03
## PC.ae.C36.4_T2     -1.206516e-02
## PC.ae.C36.5_T2     -1.529576e-02
## PC.ae.C38.0_T2      2.152600e-03
## PC.ae.C38.2_T2      6.184151e-02
## PC.ae.C38.3_T2      6.958746e-02
## PC.ae.C38.4_T2     -4.858352e-03
## PC.ae.C38.5_T2     -2.245176e-02
## PC.ae.C38.6_T2     -1.416607e-02
## PC.ae.C40.1_T2      2.618671e-02
## PC.ae.C40.2_T2      2.075903e-02
## PC.ae.C40.3_T2      6.288411e-02
## PC.ae.C40.4_T2      3.725597e-02
## PC.ae.C40.5_T2      5.241610e-02
## PC.ae.C40.6_T2     -3.626280e-03
## PC.ae.C42.1_T2      1.789557e-02
## PC.ae.C42.2_T2      1.548579e-02
## PC.ae.C42.3_T2      2.289089e-02
## PC.ae.C42.4_T2      9.786602e-03
## PC.ae.C42.5_T2      2.097016e-02
## PC.ae.C44.3_T2      4.489241e-02
## PC.ae.C44.4_T2      4.575337e-03
## PC.ae.C44.5_T2      2.689887e-04
## PC.ae.C44.6_T2      2.281602e-03
## SM..OH..C14.1_T2   -3.128806e-03
## SM..OH..C16.1_T2   -2.337047e-03
## SM..OH..C22.1_T2   -1.460378e-02
## SM..OH..C22.2_T2    1.103739e-02
## SM..OH..C24.1_T2   -1.506632e-03
## SM.C16.0_T2         -2.953034e-02
## SM.C16.1_T2         -2.176239e-02
## SM.C18.0_T2         -3.193996e-02
## SM.C18.1_T2         -2.472939e-02
## SM.C20.2_T2         3.941159e-04
## SM.C24.0_T2         -2.343616e-02
## SM.C24.1_T2        -2.394161e-02

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## MEDDM_T4	6.221103e-04
## MEDCOL_T4	-1.279878e-03
## MEDINF_T4	-1.077442e-03
## MEDHTA_T4	-7.630148e-04
## GLU_T4	-2.335043e-02
## INS_T4	6.262207e-03
## HOMA_T4	-2.913690e-03
## HBA1C_T4	-3.063996e-03
## HBA1C.mmol.mol_T4	-7.227177e-03
## PESO_T4	-5.581687e-03
## bmi_T4	-6.284647e-03
## CC_T4	-4.927665e-04
## CINT_T4	-2.542294e-03
## CAD_T4	-3.078503e-03
## TAD_T4	-1.051695e-02
## TAS_T4	-3.017671e-02
## TG_T4	-4.312488e-02
## COL_T4	-3.381578e-02
## LDL_T4	-2.136601e-02
## HDL_T4	-2.386914e-02
## VLDL_T4	-5.363939e-03
## PCR_T4	-8.291790e-02
## LEP_T4	-5.003410e-02
## ADIPO_T4	8.887434e-02
## GOT_T4	7.457260e-02
## GPT_T4	3.503376e-02
## GGT_T4	-1.079355e-02
## URICO_T4	1.216507e-02
## CREAT_T4	3.378724e-04
## UREA_T4	8.528855e-03
## HIERRO_T4	-1.662116e-02
## TRANSF_T4	-3.037808e-02
## FERR_T4	2.197843e-02
## Ile_T4	6.543982e-02
## Leu_T4	5.069705e-02
## Val_T4	2.705059e-02
## Ala_T4	3.727865e-02
## Pro_T4	3.423316e-02
## Gly_T4	8.064635e-02
## Ser_T4	1.068991e-02
## Trp_T4	-6.711082e-03
## Phe_T4	5.878263e-02
## Met_T4	-1.571397e-02
## Orn_T4	4.259717e-02
## Arg_T4	3.475179e-02
## His_T4	6.035623e-02
## Asn_T4	4.563756e-02
## Asp_T4	4.927814e-02
## Glu_T4	6.366851e-02
## Gln_T4	-3.129997e-02
## Cit_T4	1.129950e-02
## Tyr_T4	3.542797e-02
## Thr_T4	3.370749e-02
## Lys_T4	3.838280e-02

## Creatinine_T4	3.309588e-02
## Kynurenine_T4	2.106531e-02
## Putrescine_T4	-8.483349e-02
## Sarcosine_T4	-1.177845e-02
## Serotonin_T4	-4.284880e-03
## Taurine_T4	-8.950146e-03
## SDMA_T4	3.687367e-03
## C0_T4	-5.141620e-03
## C2_T4	1.784705e-02
## C3.OH_T4	7.054105e-04
## C6..C4.1.DC._T4	-9.159270e-02
## C5.DC..C6.OH._T4	2.287131e-03
## C7.DC_T4	1.676911e-02
## C8_T4	2.049179e-03
## C10_T4	4.748747e-03
## C10.1_T4	2.951269e-03
## C10.2_T4	2.915242e-04
## C14.1_T4	1.352338e-03
## C14.2_T4	8.748731e-04
## C16.1_T4	-4.802218e-03
## C16.2_T4	4.153818e-04
## C16.2.OH_T4	-4.855219e-03
## C18.1_T4	1.489636e-03
## C18.1.OH_T4	5.327879e-04
## C18.2_T4	7.599482e-04
## lysoPC.a.C16.0_T4	5.121107e-02
## lysoPC.a.C16.1_T4	4.814257e-02
## lysoPC.a.C17.0_T4	2.472207e-02
## lysoPC.a.C18.0_T4	5.044986e-02
## lysoPC.a.C18.1_T4	5.704308e-02
## lysoPC.a.C18.2_T4	5.740298e-02
## lysoPC.a.C20.3_T4	1.076509e-02
## lysoPC.a.C20.4_T4	4.276114e-02
## lysoPC.a.C24.0_T4	3.073208e-03
## lysoPC.a.C26.0_T4	2.253355e-02
## lysoPC.a.C26.1_T4	2.694835e-03
## lysoPC.a.C28.0_T4	2.424564e-02
## lysoPC.a.C28.1_T4	6.121849e-03
## PC.aa.C24.0_T4	2.262053e-02
## PC.aa.C28.1_T4	2.529209e-03
## PC.aa.C30.0_T4	4.828646e-03
## PC.aa.C32.0_T4	2.773994e-02
## PC.aa.C32.1_T4	1.250721e-02
## PC.aa.C32.3_T4	1.584270e-03
## PC.aa.C34.1_T4	-7.739405e-03
## PC.aa.C34.2_T4	-1.050548e-02
## PC.aa.C34.3_T4	-4.237827e-03
## PC.aa.C34.4_T4	-2.372289e-03
## PC.aa.C36.0_T4	1.056348e-02
## PC.aa.C36.1_T4	-4.853587e-02
## PC.aa.C36.2_T4	-3.864676e-02
## PC.aa.C36.3_T4	-4.622997e-02
## PC.aa.C36.4_T4	-1.250613e-02
## PC.aa.C36.5_T4	-3.147602e-02

```

## PC.aa.C38.0_T4      -1.271027e-03
## PC.aa.C38.1_T4      1.431643e-02
## PC.aa.C38.3_T4      -4.019321e-02
## PC.aa.C38.4_T4      -2.762959e-02
## PC.aa.C38.5_T4      -2.445981e-02
## PC.aa.C38.6_T4      -3.976156e-02
## PC.aa.C40.1_T4      3.734444e-02
## PC.aa.C40.2_T4      2.635762e-02
## PC.aa.C40.3_T4      2.113096e-02
## PC.aa.C40.4_T4      2.783414e-02
## PC.aa.C40.5_T4      -6.780948e-03
## PC.aa.C40.6_T4      -4.344218e-02
## PC.aa.C42.0_T4      6.296538e-03
## PC.aa.C42.1_T4      6.544875e-03
## PC.aa.C42.2_T4      1.236289e-02
## PC.aa.C42.4_T4      1.430619e-02
## PC.aa.C42.5_T4      6.171514e-03
## PC.aa.C42.6_T4      1.434618e-02
## PC.ae.C30.0_T4      1.476514e-02
## PC.ae.C32.1_T4      1.118356e-02
## PC.ae.C32.2_T4      6.153218e-03
## PC.ae.C34.0_T4      2.179413e-03
## PC.ae.C34.1_T4      -2.327206e-04
## PC.ae.C34.2_T4      7.819869e-03
## PC.ae.C34.3_T4      1.518003e-02
## PC.ae.C36.0_T4      4.670864e-03
## PC.ae.C36.1_T4      7.921180e-02
## PC.ae.C36.2_T4      8.465069e-03
## PC.ae.C36.3_T4      -5.425292e-03
## PC.ae.C36.4_T4      -1.336785e-02
## PC.ae.C36.5_T4      -6.016467e-03
## PC.ae.C38.0_T4      -5.724563e-03
## PC.ae.C38.2_T4      7.826863e-02
## PC.ae.C38.3_T4      6.469396e-02
## PC.ae.C38.4_T4      -2.321748e-03
## PC.ae.C38.5_T4      -1.349853e-02
## PC.ae.C38.6_T4      -1.652951e-02
## PC.ae.C40.1_T4      1.627970e-02
## PC.ae.C40.2_T4      2.914961e-02
## PC.ae.C40.3_T4      6.497773e-02
## PC.ae.C40.4_T4      4.842871e-02
## PC.ae.C40.5_T4      6.573000e-02
## PC.ae.C40.6_T4      -5.342771e-03
## PC.ae.C42.1_T4      1.395189e-02
## PC.ae.C42.2_T4      1.026979e-02
## PC.ae.C42.3_T4      2.066689e-02
## PC.ae.C42.4_T4      1.111241e-02
## PC.ae.C42.5_T4      3.459445e-02
## PC.ae.C44.3_T4      9.049013e-03
## PC.ae.C44.4_T4      4.294970e-03
## PC.ae.C44.5_T4      3.222760e-03
## PC.ae.C44.6_T4      6.820969e-03
## SM..OH..C14.1_T4    6.848421e-03
## SM..OH..C16.1_T4    3.716071e-03

```

```

## SM..OH..C22.1_T4 -4.151286e-02
## SM..OH..C22.2_T4 -1.095592e-03
## SM..OH..C24.1_T4 -2.392746e-03
## SM.C16.0_T4 -1.278910e-02
## SM.C16.1_T4 1.136336e-02
## SM.C18.0_T4 3.942216e-03
## SM.C18.1_T4 1.761095e-02
## SM.C20.2_T4 1.036226e-03
## SM.C24.0_T4 -5.001151e-02
## SM.C24.1_T4 -2.457752e-02
## MEDDM_T5 1.541726e-03
## MEDCOL_T5 1.541726e-03
## MEDINF_T5 3.187324e-04
## MEDHTA_T5 1.631254e-03
## GLU_T5 -5.527453e-02
## INS_T5 -4.071259e-03
## HOMA_T5 -6.458594e-03
## HBA1C_T5 -4.985002e-03
## HBA1C.mmol.mol_T5 -1.389622e-02
## PESO_T5 -2.179200e-02
## bmi_T5 -2.150532e-02
## CC_T5 7.249484e-05
## CINT_T5 -2.481676e-02
## CAD_T5 -2.543660e-02
## TAD_T5 -9.652439e-03
## TAS_T5 -5.731885e-04
## TG_T5 -5.063651e-02
## COL_T5 -6.630115e-02
## LDL_T5 -5.459825e-02
## HDL_T5 -2.128401e-02
## VLDL_T5 -3.610009e-02
## PCR_T5 -5.089461e-02
## LEP_T5 -4.461574e-02
## ADIPO_T5 8.287738e-02
## GOT_T5 7.650004e-02
## GPT_T5 3.879732e-02
## GGT_T5 1.371071e-02
## URICO_T5 1.750099e-03
## CREAT_T5 1.526487e-04
## UREA_T5 1.438625e-03
## HIERRO_T5 2.410971e-03
## TRANSF_T5 -4.065783e-02
## FERR_T5 7.311036e-03
## Ile_T5 6.129890e-02
## Leu_T5 6.038558e-02
## Val_T5 4.059724e-02
## Ala_T5 3.978706e-02
## Pro_T5 6.525215e-02
## Gly_T5 6.657912e-02
## Ser_T5 7.261378e-02
## Trp_T5 8.324534e-03
## Phe_T5 7.489163e-02
## Met_T5 -1.800008e-02
## Orn_T5 1.019141e-01

```

## Arg_T5	4.046660e-02
## His_T5	8.287990e-02
## Asn_T5	8.112528e-02
## Asp_T5	9.374592e-02
## Glu_T5	5.834101e-02
## Gln_T5	4.361097e-02
## Cit_T5	1.036185e-01
## Tyr_T5	7.664697e-02
## Thr_T5	9.804510e-03
## Lys_T5	1.077291e-02
## Creatinine_T5	2.462916e-02
## Kynurenine_T5	-8.851449e-03
## Putrescine_T5	-2.499126e-02
## Sarcosine_T5	-1.309200e-03
## Serotonin_T5	5.082634e-03
## Taurine_T5	4.741507e-02
## SDMA_T5	3.343762e-03
## C0_T5	3.783909e-02
## C2_T5	1.585730e-02
## C3.OH_T5	7.224460e-04
## C6..C4.1.DC._T5	-1.463326e-02
## C5.DC..C6.OH._T5	3.343447e-03
## C7.DC_T5	-9.757712e-03
## C8_T5	1.766735e-03
## C10_T5	3.071888e-03
## C10.1_T5	6.500279e-02
## C10.2_T5	6.744042e-04
## C14.1_T5	1.246695e-03
## C14.2_T5	1.016741e-03
## C16.1_T5	5.211026e-04
## C16.2_T5	5.672111e-04
## C16.2.OH_T5	1.053338e-02
## C18.1_T5	1.956490e-03
## C18.1.OH_T5	4.442292e-04
## C18.2_T5	9.834848e-04
## lysoPC.a.C16.0_T5	7.080223e-02
## lysoPC.a.C16.1_T5	6.900519e-02
## lysoPC.a.C17.0_T5	2.487366e-02
## lysoPC.a.C18.0_T5	7.257095e-02
## lysoPC.a.C18.1_T5	8.201356e-02
## lysoPC.a.C18.2_T5	7.123098e-02
## lysoPC.a.C20.3_T5	3.758766e-02
## lysoPC.a.C20.4_T5	5.087245e-02
## lysoPC.a.C24.0_T5	5.029244e-03
## lysoPC.a.C26.0_T5	1.097507e-02
## lysoPC.a.C26.1_T5	3.042777e-03
## lysoPC.a.C28.0_T5	3.751754e-02
## lysoPC.a.C28.1_T5	4.852821e-03
## PC.aa.C24.0_T5	1.902667e-02
## PC.aa.C28.1_T5	3.609349e-03
## PC.aa.C30.0_T5	2.100091e-03
## PC.aa.C32.0_T5	4.776583e-02
## PC.aa.C32.1_T5	3.429672e-02
## PC.aa.C32.3_T5	1.614148e-03

## PC.aa.C34.1_T5	2.761893e-02
## PC.aa.C34.2_T5	1.135867e-02
## PC.aa.C34.3_T5	1.254723e-02
## PC.aa.C34.4_T5	3.564734e-04
## PC.aa.C36.0_T5	1.467368e-02
## PC.aa.C36.1_T5	4.375246e-02
## PC.aa.C36.2_T5	1.135660e-02
## PC.aa.C36.3_T5	7.302783e-03
## PC.aa.C36.4_T5	-4.911345e-03
## PC.aa.C36.5_T5	-2.951559e-02
## PC.aa.C38.0_T5	7.425625e-03
## PC.aa.C38.1_T5	2.500377e-02
## PC.aa.C38.3_T5	8.581703e-03
## PC.aa.C38.4_T5	-1.115337e-03
## PC.aa.C38.5_T5	-2.278655e-03
## PC.aa.C38.6_T5	-3.171722e-02
## PC.aa.C40.1_T5	5.057646e-02
## PC.aa.C40.2_T5	2.390378e-02
## PC.aa.C40.3_T5	2.484305e-02
## PC.aa.C40.4_T5	2.942227e-02
## PC.aa.C40.5_T5	1.472309e-02
## PC.aa.C40.6_T5	-2.198015e-02
## PC.aa.C42.0_T5	5.582795e-03
## PC.aa.C42.1_T5	4.445405e-03
## PC.aa.C42.2_T5	1.134628e-02
## PC.aa.C42.4_T5	1.436200e-02
## PC.aa.C42.5_T5	8.337280e-03
## PC.aa.C42.6_T5	3.059907e-02
## PC.ae.C30.0_T5	4.123809e-02
## PC.ae.C32.1_T5	1.464275e-02
## PC.ae.C32.2_T5	6.155085e-03
## PC.ae.C34.0_T5	4.204792e-03
## PC.ae.C34.1_T5	1.500118e-02
## PC.ae.C34.2_T5	2.976525e-02
## PC.ae.C34.3_T5	3.032804e-02
## PC.ae.C36.0_T5	1.012804e-02
## PC.ae.C36.1_T5	8.529050e-02
## PC.ae.C36.2_T5	3.312514e-02
## PC.ae.C36.3_T5	1.091268e-02
## PC.ae.C36.4_T5	1.211299e-02
## PC.ae.C36.5_T5	9.455720e-03
## PC.ae.C38.0_T5	-3.775928e-04
## PC.ae.C38.2_T5	7.961989e-02
## PC.ae.C38.3_T5	5.944123e-02
## PC.ae.C38.4_T5	2.808046e-03
## PC.ae.C38.5_T5	-5.304074e-04
## PC.ae.C38.6_T5	-5.422127e-03
## PC.ae.C40.1_T5	1.752307e-02
## PC.ae.C40.2_T5	3.109212e-02
## PC.ae.C40.3_T5	6.101128e-02
## PC.ae.C40.4_T5	4.657417e-02
## PC.ae.C40.5_T5	5.827443e-02
## PC.ae.C40.6_T5	-1.353030e-05
## PC.ae.C42.1_T5	1.483743e-02

```
## PC.ae.C42.2_T5      1.296022e-02
## PC.ae.C42.3_T5      2.042069e-02
## PC.ae.C42.4_T5      1.466883e-02
## PC.ae.C42.5_T5      3.683359e-02
## PC.ae.C44.3_T5      9.591082e-03
## PC.ae.C44.4_T5      5.611252e-03
## PC.ae.C44.5_T5      2.671604e-03
## PC.ae.C44.6_T5      6.367906e-03
## SM..OH..C14.1_T5    5.134371e-03
## SM..OH..C16.1_T5    1.238687e-03
## SM..OH..C22.1_T5    -2.926439e-02
## SM..OH..C22.2_T5    -3.646292e-03
## SM..OH..C24.1_T5    -1.647480e-03
## SM.C16.0_T5         9.260947e-03
## SM.C16.1_T5         2.088881e-02
## SM.C18.0_T5         2.338297e-03
## SM.C18.1_T5         1.268094e-02
## SM.C20.2_T5         2.555965e-04
## SM.C24.0_T5         -2.272492e-02
## SM.C24.1_T5         3.476208e-03
```

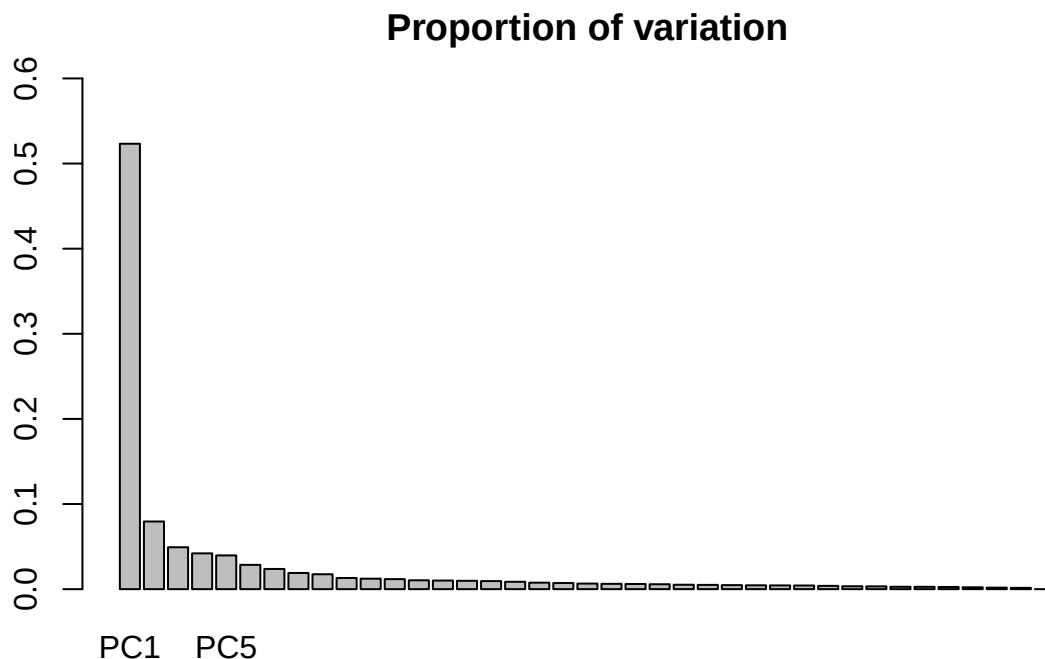
```
summary(PCaseMA)
```

```
## Importance of first k=5 (out of 39) components:
##              PC1      PC2      PC3      PC4      PC5
## Standard deviation    18.9326  7.37793  5.80678  5.36651  5.21288
## Proportion of Variance  0.5233  0.07947  0.04922  0.04204  0.03967
## Cumulative Proportion  0.5233  0.60274  0.65196  0.69401  0.73368
```

L'anàlisi no és molt concloent. Per obtenir un 70% de la variancia ens calen 5 components, el que és molt. Entenc que hi ha massa variables en joc i segurament caldria reduir abans d'alguna altra manera. Segurament hi ha explicacions biològiques que estaria bé aplicar, però a mi se m'escapen ja que no comprenc totes les variable i per tant, acoto aquí.

Tot i això, per veure com evoluciona l'explicació de la varianza faig un gràfic de barres amb una nova PCA que porta tots els components

```
barplot(summary(PCaseMA)$importance[2,],ylim=c(0,0.6))
title(main="Proportion of variation",line=1)
```



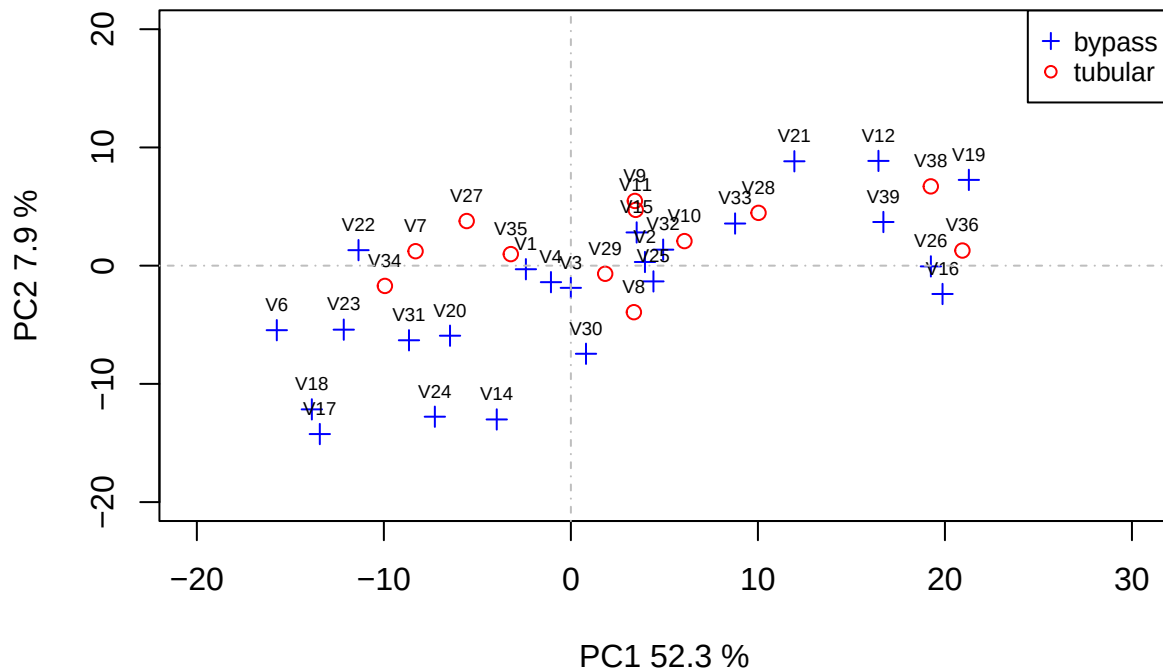
S'pareia força clar com la primera aporta força i les següents van afegint poc. Tot i això, podríem dir que fins a la cinquena sembla interessant agafarles.

Grafiquem ara les primeres components i ho fem en relació al tipus de cirurgia que és el que cerca l'article original. De moment, dues

```
# carreguem les dades pel gràfic
cirugia <- as.factor(rowData(seMA)$SURGERY)
cirugian <- as.numeric(cirugia)
# Amb això hem passat les dades de cirurgia amb 1: by pass i 2: tubular
loads <- round(PCaseMA$sdev^2/sum(PCaseMA$sdev^2)*100,1)
xlab <- c(paste("PC1",loads[1],"%"))
ylab <- c(paste("PC2",loads[2],"%"))
# grafiquem
plot(PCaseMA$x[,1:2],
     xlab=xlab,
     ylab=ylab,
     pch = c(3,21)[cirugia],
     col=c("blue","red")[cirugia],
     main = "Principal components (PCA)",
     ylim=c(-20,20),
     xlim=c(-20,30))
names <- rownames(PCaseMA$x)
text(PCaseMA$x[,1],PCaseMA$x[,2],names, pos=3, cex=.6)
abline(h=0,v=0,lty=4,col="gray")
legend("topright",
     pch=c(3,21),
     col=c("blue","red"),
```

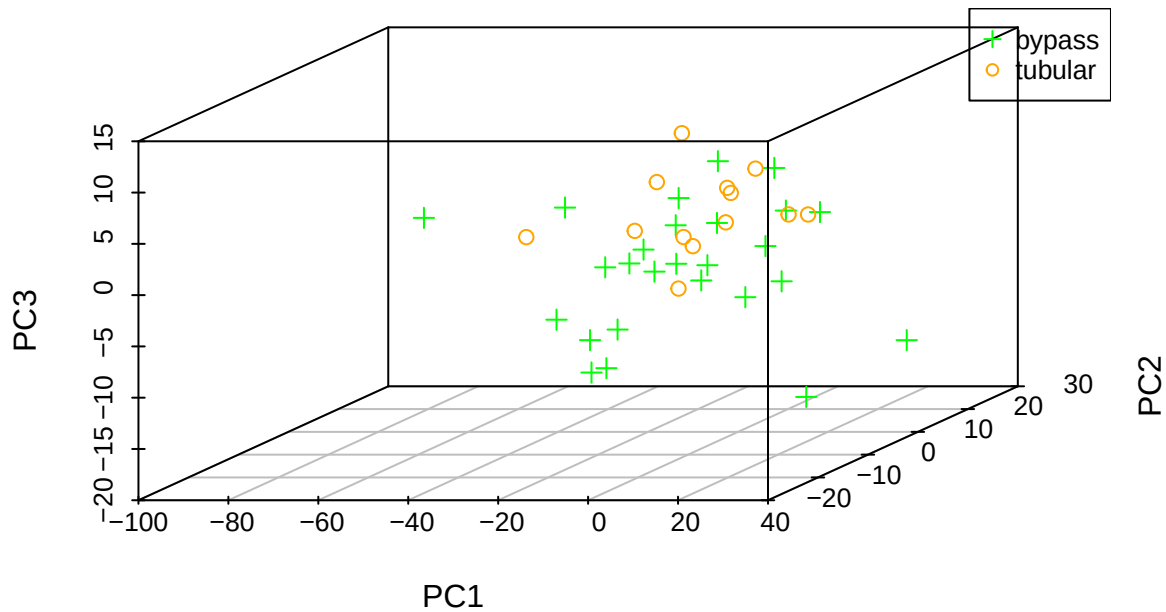
```
cex=0.8,
legend=c("bypass", "tubular"))
```

Principal components (PCA)



Sembla que les dues components ens separen els dos tipus de cirurgia, el que implicaria una resposta diferent dels metabolits a la cirurgia. Provem si amb 3 dimensions millora

```
gr3Ddata <- as.data.frame(PCaseMA$x)
gr3Ddata$cirurgia <- cirugian
library(scatterplot3d)
# scatterplot3d(PCaseMA$x[,1], PCaseMA$x[,2], PCaseMA$x[,3],
scatterplot3d(gr3Ddata[,1], gr3Ddata[,2], gr3Ddata[,3],
              xlab="PC1", ylab="PC2", zlab="PC3",
              color = ifelse(gr3Ddata$cirurgia == "1", "green", "orange"),
              pch = ifelse(gr3Ddata$cirurgia == "1", 3, 21))
legend("topright",
      pch=c(3,21),
      col=c("green", "orange"),
      cex=0.8,
      legend=c("bypass", "tubular"))
```

Sembla que en 3 dimensions encara es veu més l'efecte de la cirurgia

Amb plotly ho podem veure més fàcil al rotar el gràfic. El possem per fer el html, però no per fer el pdf, ja que no funciona amb el latex.

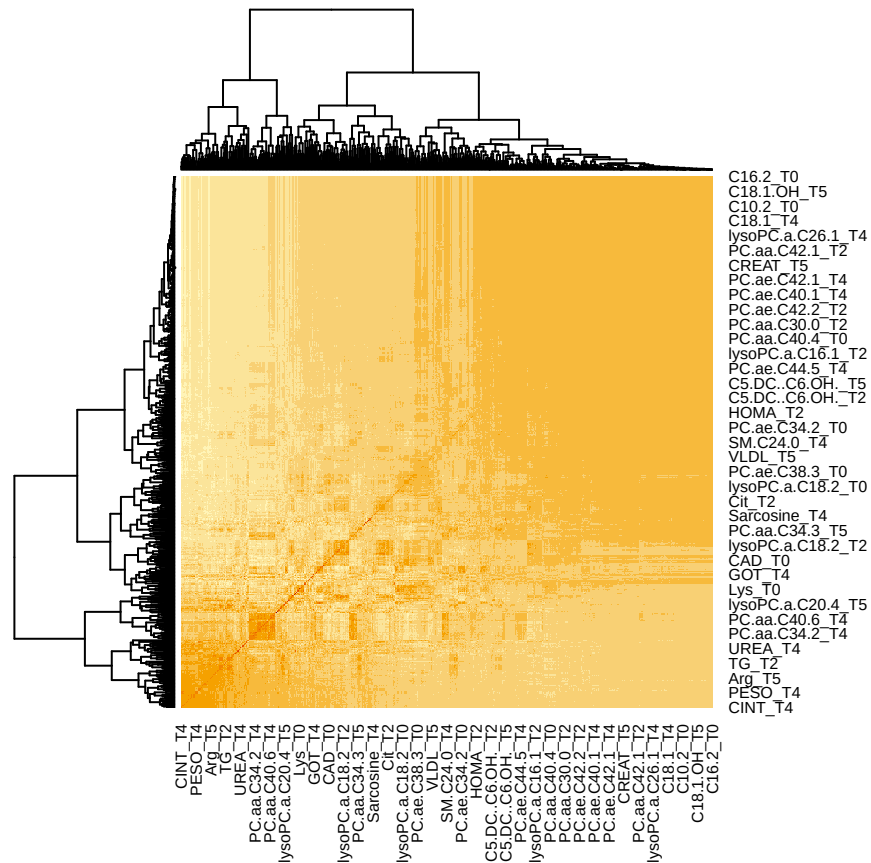
```
#library(plotly)
#fig <- plot_ly(gr3Ddata,
#               x = ~gr3Ddata[,1], y = ~gr3Ddata[,2], z = ~gr3Ddata[,3],
#               color = ~cirugia,
#               type = 'scatter3d',
#               mode = 'markers',
#               marker = list(size = 5))

#fig <- fig %>%
#  layout(title = 'PCA en 3D',
#         scene = list(
#           xaxis = list(title = 'PC1'),
#           yaxis = list(title = 'PC2'),
#           zaxis = list(title = 'PC3')
#         ))
#fig
```

7. Cerca de patrons

Ho farem amb un heatmap de la matriu de correlacions que permetria veure totes les relacions

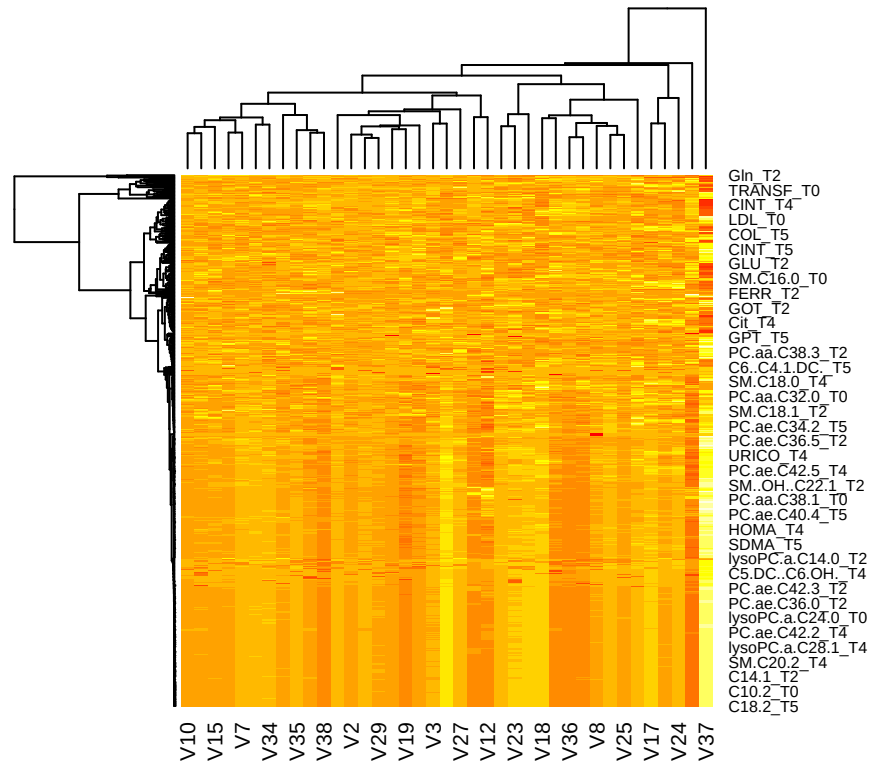
```
cor_matrix <- cor(na.omit(assay(seMANormalized)))
heatmap(cor_matrix)
```



Es veuen dos grups de variables més correlacionades, però en general les correlacions són debils.

Ara fem un altre heatmap que calcula distancies entre les variables i les mostres

```
dist_matrix <- t(assay(seMANormalized))
heatmap(dist_matrix, col=heat.colors(16))
```



Hi ha tantes mostres i variables que no es poden veure be totes les relacions, però sembla que hi ha un patró clar de proximitat en les primeres variables del gràfic i algunes que estan a molta distància cap a baix.